Public User Services & Technology: a Primer for Librarians

PUBLIC USER SERVICES & TECHNOLOGY: A PRIMER FOR LIBRARIANS

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ABOUT THIS BOOK

Public User Services & Technology: A Primer for Librarians was developed as an Open Educational Resource through an OpenHawks grant at the University of Iowa Libraries. The book was created within a graduate course on Public User Services & Technology, with chapters written by enrolled students and edited by course instructor Micah Bateman in collaboration with graduate assistant and co-editor Sera-Ann Hargrove.

Each chapter is designed as a concise overview of a core topic relevant to contemporary public-facing library work, particularly as it intersects with technology, digital resources, and online services. Chapters were developed to align with individual course modules, introducing foundational concepts, professional standards, and common practices that beginning librarians and library workers are expected to encounter in the field.

While the book originated in a classroom setting, it is intended to be useful beyond that context. The chapters are written to stand alone and may be read selectively, making the book suitable for self-directed learning, staff training, or use in other library and information science courses. The intended audience includes MLIS students, early-career librarians, paraprofessionals, and library workers seeking a practical introduction to public user services and technology.

The book is organized in two parts. Part I introduces key areas of public user services, including library technology, digital literacy, accessibility, user experience, reference services, search strategies, and technology management. Part II builds on this foundation by examining how differences in access, skills, and resources shape user interactions with technology across a range of populations and contexts. Taken together, the chapters provide readers with a shared vocabulary and conceptual framework for understanding how libraries support patrons' use of technology in twenty-first-century information environments.

Note on Format and Media

This book is available in multiple formats. Some chapters include embedded or interactive media elements (such as videos or external tools) in the web-based version. These interactive elements may not display in PDF or print versions of the text. Readers using a static format are encouraged to consult the online version of the book for access to all media content.

INTRODUCTION

Olivia Backes; Schuyler Fynaardt; and Robyn Gardner

Instructor Framing

This course—and the book that emerged from it—is grounded in a particular understanding of public librarianship, shaped in part by Jessamyn West's *Without a Net*. West argues that libraries increasingly function as a kind of civic help desk: a place where people arrive with immediate, often practical problems related to technology, access, and information systems, and where librarians respond through patient, situated assistance.

From this perspective, public user services are less about mastering discrete tools than about developing the capacity to diagnose needs, adapt explanations, and work collaboratively with patrons in moments of uncertainty. The digital divide appears throughout this book not as an abstract problem to be solved once and for all, but as one of the recurring conditions that shape help desk interactions in real library settings.

The chapters that follow reflect this orientation. Written by students in a course designed around public-facing service work, they emphasize practical scenarios, service frameworks, and the everyday labor of supporting patrons as they navigate technology, resources, and institutional systems.

Introduction

In her book *Without a Net*, Jessamyn West defines the digital divide as "the gap between people who can easily use and access technology, and those who cannot" (West, 2011, p. xxiv). In the twenty-first century, librarians are in a unique position to promote digital literacy; while libraries have always been centers for learning, the digital age brings with it a responsibility to ensure all patrons' ability to use technology and access digital resources. Consequently, when exploring how best to develop technology instruction and digital literacy-based library programming, there are a variety of factors to consider, including relationships with patrons, complications with bridging the digital divide, and access to library e-resources.

An important element of librarianship is building relationships with patrons in order to encourage curiosity, promote lifelong learning, and support access to information. In order to help patrons keep an open mind, librarians should work to reframe technology as exploration, rather than judgment. To assist with this, understanding patrons' prior knowledge and their goals for using technology is key to developing effective programming (West, 2011, p. 84). Likewise, digital literacy instruction in diverse communities can be tailored to address cultural considerations related to technology use and can be offered in multiple languages (Inamdar,

n.d., p. 3). Considering the people and populations we're serving is critical for ensuring all people have access to information and technology instruction.

As librarians aim to promote digital literacy, they must work through this divide and its complexities to bridge users to their technology. There's no shortage of complications that can occur during this process, such as:

- Negative attitudes surrounding technology
- Perpetual technology catch-up
- Funding
- Difficulty identifying underserved groups
- Inconsistency of technological resources between libraries
- Lack of accessibility
- Difficulty reaching patrons without technology

Furthermore, libraries may be the only place some people can go online. These patrons often use library technology out of necessity, which may increase emotional tensions. In the pursuit of teaching and providing technology, libraries should "be mindful of mission, limitation, and the community context in which we operate" to identify problems and solutions unique to each library (West, 2011, p. 75). Working through the complexities of the digital divide is an essential part of librarians' work.

Finally, a key mission for libraries is providing legitimate access that keeps people reading, a goal now attained with e-resources. This task is complicated by the fact that "a majority of library digital collections non-users say they don't know how to access digital collections" (Berens & Noorda, 2023). This lack of digital literacy effectively creates a major barrier to use, leaving the library's substantial investment in electronic materials underutilized and potentially widening the digital divide within the community. Therefore, librarians are actively working not only to expand access (by offering Wi-Fi and devices) but also to bridge this knowledge gap. They connect with the community through educational programming––offering everything from one-on-one tech help sessions to formal workshops––that teaches patrons how to navigate platforms, download materials, and manage various digital formats. This proactive, hands-on approach ensures that the library fulfills its mission by transforming mere availability into meaningful access and skill-based empowerment.

In addition to building relationships with patrons, navigating complications with the digital divide, and promoting access to digital resources, the following chapters will provide more insight into how best to ensure all people have access to the knowledge and tools needed to participate in twenty-first-century life.

Works Cited

Berens, K. I., & Noorda, R. (2023). Gen Z and Millennials How They Use Public Libraries and Identify through Media Use. *PDXScholar.* pdxscholar.library.pdx.edu/eng_fac/135/.

Inamdar, S. (n.d). The Role of Libraries in Promoting Digital Literacy in the 21st Century. *SSRN.* papers.ssrn.com/sol3/papers.cfm?abstract_id=4437672.

West, J. C. (2011). *Without a Net: Librarians Bridging the Digital Divide*. Libraries Unlimited.

PART I
PUBLIC USER SERVICES

1.

LIBRARY TECHNOLOGY

Kelsey Behnke; Laura Hughes Donaldson; and Liz Shatek

Learning Objectives

- Familiarize yourself with current library technology, its uses, and impacts.
- Familiarize yourself with relevant technology trends that may impact librarianship in the future.

Current Library Technologies

Libraries have long served as a community hub for technology, but recent developments in technology applications have made this role more vital than ever. With almost every aspect of life having an online counterpart—books, job applications, education, and banking—the need for people to access technology has become essential.

According to the Public Library Association's (PLA) *Public Library Technology Survey Summary Report*, most library technologies fall into one of three categories: on-site, off-site, and online (Public Library Association, 2024).

On-site technologies that appear in most (>50%) public libraries include copy machines, scanners, color printers, fax machines, wireless or mobile printing, laptops, early learning devices, and smart objects for STEAM.

Off-site technologies, though less common, often include items available for checkout such as internet hotspots, laptops, and recreational technology.

Online resources widely available through libraries include e-books and audiobooks, streaming media, job and employment resources, health and genealogy tools, and language learning platforms.

Uses and Impacts

By offering access to these technologies, libraries empower individuals who might otherwise lack the means to engage in essential digital activities, such as applying for jobs, creating resumes, completing schoolwork, researching health information, or connecting with others online. This access reduces financial barriers, allowing patrons to allocate limited funds elsewhere while still participating fully in modern life.

Libraries also provide a safe environment for patrons to explore and learn about new technologies. By providing workshops on topics such as digital literacy, online safety, and device use, libraries ensure that patrons have the skills and access to use technology effectively, thus acting as both a provider and an educator in promoting digital equity across communities.

Looking Ahead: Technology Trends

As technology continues to evolve, libraries must anticipate and adapt to emerging trends. The American Library Association highlights several developments likely to influence librarianship in the coming years, including artificial intelligence (AI), connected toys, drones, haptic technology, and virtual reality (American Library Association, n.d.). These innovations present opportunities to enhance services and engagement but also raise new questions about ethics, accessibility, and information reliability.

AI, in particular, has become increasingly popular in the last few years, in an abundance of forms and with few regulations. There is a good chance that libraries will encounter it at some point if they have not already, whether that is through patrons using AI tools, AI-generated content within databases, or new automated systems for cataloging and research support. As AI becomes more integrated into everyday life, libraries will need to evaluate its implications and develop thoughtful policies around its use to maintain trust, accuracy, and inclusivity.

Conclusion

Libraries have always adapted to meet the evolving needs of their communities. Today, their technological resources ensure that everyone has the opportunity to participate in the digital world. As technology continues to advance, libraries will remain essential in providing equitable access, fostering digital literacy, and guiding communities through the ever-changing landscape of information and innovation.

Works Cited

American Library Association. (n.d.). *Trends | Center for the Future of Libraries*. ALA. https://www.ala.org/future/trends

Public Library Association. (2024). *2023 Public Library Technology Survey: Summary report.*

American Library Association. https://www.ala.org/sites/default/files/2024-07/PLA_Tech_Survey_Report_2024.pdf

2.

DIGITAL LITERACY

Linda Brown; Sam Plank; and Amber Riedemann

Learning Objectives

- Enumerate objectives for digital literacy standards
- Reflect on digital literacy standards expectations and implementations

With technology at the forefront of our daily lives, librarians are expected to be familiar with a variety of digital resources so they can fully aid patrons. With so many types of resources, it can be difficult for professionals to determine which skills are essential to know. By learning the basics of digital literacy, librarians can become more confident with their skillsets and assist readers with their use of technology.

According to the American Library Association (ALA), digital literacy is "the ability to use information and communication technologies to find, evaluate, create, and communicate information" (American Library Association [ALA], n.d.). This involves the skills needed to locate, and analyze digital tools (ALA, n.d.). Digital literacy is essential because our world relies on being able to find and use technology. Without this skill, disinformation can spread quickly through misunderstandings (Radovanovic, 2025).

As mentioned above, technology is integrated in life in the world today. Librarianship is very connected to digital literacy as communities connect with the libraries of today. With the reality of technology being weaved in society, librarians often have a significant role in encouraging digital literacy, as communities exist within the world.

According to the ALA, sub-competencies that embody the connection between librarianship and digital literacy are "Identify appropriate technologies and uses that support access to and delivery of library services and resources [...] Evaluate existing and emerging technologies and their impact on library services and resources in terms of accessibility, practicality, sustainability, and effectiveness" (American Library Association, 2023). This is reflected today as librarians assist patrons with digital learning and competency, as well as equipping digital literacy throughout their careers. Librarians often teach digital literacy classes, as reflected at the Iowa City Public Library where tech classes are provided to equip patrons across all skill levels with the opportunity to learn more (Iowa City Public Library, n.d.). The standards stated above acknowledge the

role librarians have in digital literacy, creating space for information access, instructional support, resource curation, technology integration, and community engagement.

In recognition of the importance of being digitally literate, many schools are including instruction in basic computer skills and digital citizenship in cooperation with school librarians (American Association of School Librarians, 2018; Utah State Board of Education, 2022). Digital literacy learning serves as an ever more vital foundation for information literacy as more information is accessed via the internet or by other digital technologies. For higher education and adult learning settings––academic and public libraries, respectively––digital literacy is developed in sets of nuanced "knowledge practices" and "dispositions" rather than discrete skills, as laid out by the Association of College and Research Libraries (ACRL) (Association of College and Research Libraries, 2015). Each of these "frames", as the ACRL defines them, are embedded within the daily work of librarianship and information work in general.

Key Takeaway

Digital literacy is a foundational competency for contemporary librarianship, encompassing not only technical skills but also the ability to evaluate, communicate, and teach information in digital environments. Librarians support patrons' use of technology by aligning instruction with professional standards and adapting to changing modes of information access.

Works Cited

American Library Association. (n.d.). *Digital literacy*. ALA Literacy Clearinghouse. https://literacy.ala.org/digital-literacy/

American Association of School Librarians. (2018). *School library evaluation checklist*. https://standards.aasl.org/wp-content/uploads/2018/10/180921-aasl-standards-evaluation-checklist-color.pdf

American Library Association. (2023, January 28). *ALA's Core Competences of Librarianship*. https://www.ala.org/sites/default/files/educationcareers/content/2022%20ALA%20Core%20Competences%20of%20Librarianship_FINAL.pdf

Association of College and Research Libraries. (2015, February 2). *Framework for information literacy for higher education*. American Library Association. https://www.ala.org/sites/default/files/acrl/content/issues/infolit/framework1.pdf

Iowa City Public Library. (n.d.). *Tech classes*. Iowa City Public Library. https://www.icpl.org/events/series/tech-classes

Radovanovic, D. (2025). In *Digital literacy and inclusion: Stories, platforms, communities* (pp. vii–vii). Forward, Springer Nature Switzerland AG.

Utah State Board of Education. (2022, October). *Digital literacy strands and standards*. https://www.utah.gov/pmn/files/952815.pdf

3.

DIGITAL DIVIDE

Kaitlin McAbee; Hannah Miraglia; and Rachel Winey

Learning Objectives

- Recognize the key terms in context
- Learn what causes the issues of digital inaccessibility
- Understand the role of libraries in addressing the digital divide

Key Terms

- **Digital divide:** uneven distribution in access, use, and skills
- **Digital inclusion:** strategies to ensure access, devices, and education for all
- **Digital equity:** condition in which everyone has the ability to engage fully
- **Digital literacy:** knowledge and skills for digital participation

Digital Divide

The *digital divide* has become a serious problem as access to the internet has become a necessity in contemporary society rather than just being a luxury. Librarians must help discover a way to bridge the divide over time, expanding on their understanding of the issue from access to literacy.

Access was the main topic of early discussions of the digital divide, seen as a split between those who could connect and those who couldn't. King (2021) named availability and affordability as the main two obstacles to access. Geographically, many rural places just couldn't have high-speed networks nor afford broadband services. Sometimes costs would even make access impossible in urban centers. Thus the solution might seem to be expanding broadband infrastructure and subsidizing costs to create *digital inclusion*. However, even if libraries can help bridge the access gap, that alone won't fix the problem found in the new digital divide.

Despite an estimated 80% of Americans subscribed to home broadband internet (Gelles-Watnick, 2024), digital disparity persists in new forms. The new digital divide refers to gaps in digital access *and* in use and competency. A 2024 study challenges the notion that infrastructure alone will erase the digital divide (Pereira et al., 2024). Some areas with strong infrastructure still demonstrate low use. Findings also correlate lower

income and education levels to lower technology use and literacy even when access is present (Pereira et al., 2024). *Digital equity* is necessary to ensure that everyone can participate in and benefit from digital activity in areas like civic engagement, employment, connectedness, education, and essential services. Information professionals play a key role in addressing the new digital divide by facilitating access, education, and support.

Many libraries offer services that help address some of the issues around the digital divide and digital equity. According to a survey done by the Public Library Association (2024), 67% of city and suburban libraries surveyed offered internet hotspots for patrons to checkout while only 39% of town and rural libraries offered hotspots as well. These hotspots can help narrow the digital divide by helping people who cannot access the internet in other ways, either due to geographic location or due to the cost. Libraries also help with digital equity by providing training related to different aspects of *digital literacy*. According to the same survey, 92% of libraries provide formal and/or informal training on general computer skills (e.g., using mouse and keyboard, word processing) and internet use (e.g., searching, social media, online safety) (Public Library Association, 2024).

Closing the digital divide and inequity requires more than just expanding infrastructure, it demands sustained attention. Libraries remain essential partners in the effort to provide the access, education, and community support that empower individuals to fully participate in a digital world, now and in the future.

To read more about the digital divide, see Part II of this textbook, where we explore the digital divide and how it effects age, disability, immigrants, Indigeneity, race, rurality, and socioeconomics.

Works Cited

Gelles-Watnick, R. (2024, Jan. 31). American use of mobile technology and home broadband. *Pew Research Center.* https://www.pewresearch.org/internet/2024/01/31/americans-use-of-mobile-technology-and-home-broadband/

King, D. L. (2021). The digital divide. *Library Technology Reports, 57*(2), 24–28. American Library Association. https://alatechsource.org

Pereira, M., Greenstein, S., Sadun, R., Tambe, P., Ronchi Darre, L., Glazer, T., Kim, A., Dodhia, R., & Lavista Ferres, J. (2024). The new digital divide (NBER Working Paper No. 32932). *National Bureau of Economic Research.* https://www.nber.org/papers/w32932

Public Library Association. (2024). 2023 Public Library Technology Survey: Summary Report. *Chicago: Public Library Association.* https://www.ala.org/sites/default/files/2024-07/PLA_Tech_Survey_Report_2024.pdf

4.

WEB ACCESSIBILITY

Sarah Ramsdell; Vashti Torres; and Luke Vidic

Learning Objectives

- Enumerate methods for making web documents and websites more accessible
- Locate tools and standards for optimizing web accessibility

Introduction

As of July 2025, more than 67% of the global population have accessed the internet (Petrosyan, 2025). Around 1.3 billion users have a disability that requires some kind of assistive tool or technology to access online content (Milbergs, 2023). Still, less than 2% of websites make such accommodations, leading to low-contrast or missing alternative text, empty links or buttons, and no input form labels (Milbergs, 2023). Version 2.2 of The Web Content Accessibility Guidelines (WCAG) offers guidelines that make online content more accessible to more people, including those with disabilities (W3C, 2025). While incapable of accommodating every disability, the guidelines promote inclusivity by addressing limitations resulting from blindness and low vision, deafness and hearing loss, limited movement, speech disabilities, photosensitivity, as well as some learning disabilities and cognitive limitations.

WCAG Guidelines Overview

The WCAG provides accessibility standards for websites. Entities who use these guidelines include developers of web content, developers of web authoring tools, and developers of web tool accessibility evaluators, as well as others who use it on a technical level. WCAG 2.1 and 2.2 were updated as the internet changed and grew. There are thirteen guidelines that fit under four different principles: perceivable, operable, understandable, and robust (WAI, 2025).



Figure 1. Graphic depicting the four principles of web accessibility: perceivable, operable, understandable, and robust.

Guidelines alone may not be enough to guarantee accessibility but they can give developers an ideal to strive toward (Georgakas, 2023).

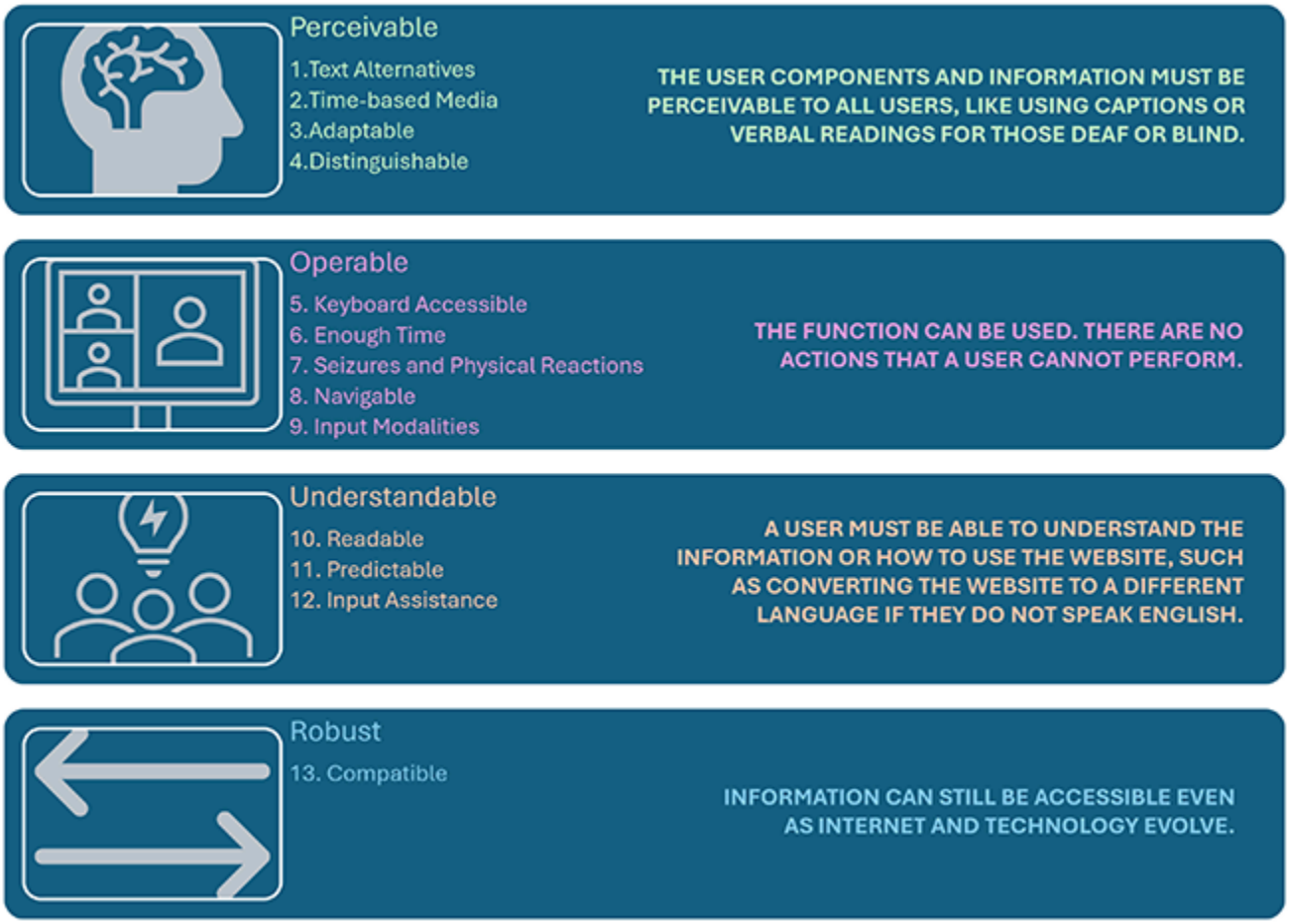


Figure 2. Infographic of the thirteen guidelines within the four principles of web accessibility.

Key Components of Web Accessibility

According to the Web Accessibility Initiative (WIA), the accessibility of web content relies on several factors. Developers create content through web-authoring software, reviewing the accessibility of their work using evaluation tools; on the other side of the screen, users rely on their own knowledge, abilities, and available assistive tools to interact with user agents (web browsers and media players) to access content. Each component in both pathways affects the accessibility of web content. If one component in either pathway fails to meet the user's needs, inefficient workarounds may be created, and other components may work harder to meet the user's needs.

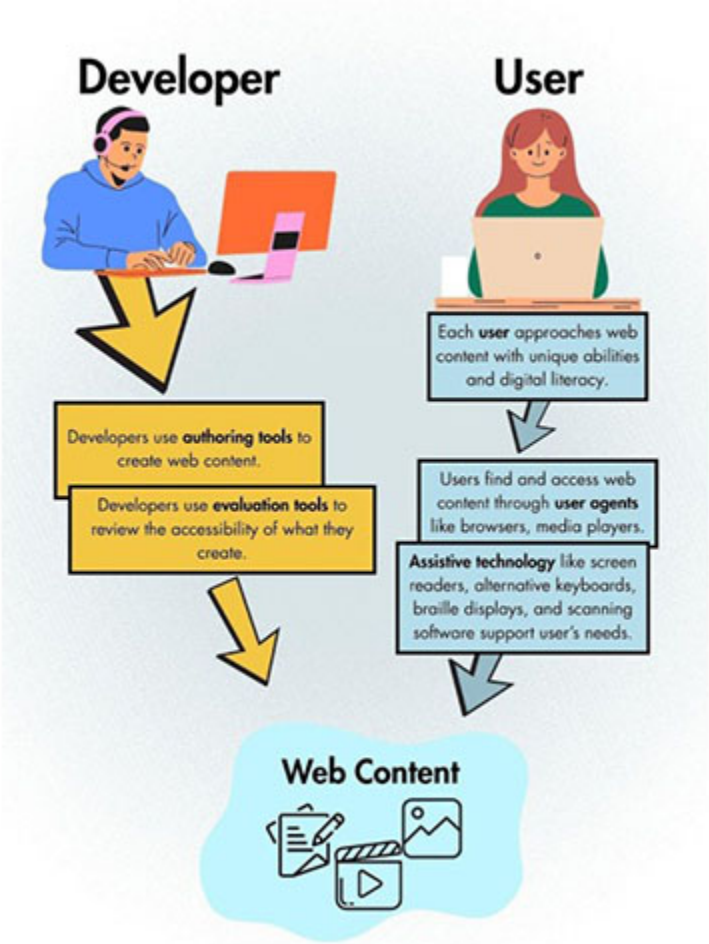


Figure 3. Graphic depicting how developers and users interact with web content.

Real-World Impact

Private websites are not required by law to comply with WCAG, though websites do need to be accessible, a fact which is often cited by the Americans with Disabilities Act (ADA) in cases regarding a failure to do so (ADA.gov, 2024; Bureau of Internet Accessibility, 2019). By complying with guidelines, business and commercial websites increase audience reach, improve experiences, optimize sites for search engines, enhance brand inclusivity, and avoid potential lawsuits. For individuals, accessible websites could unlock opportunities for employment, education, and participation in digital society, all of which remove barriers leading to social isolation and economic disadvantage (AllAccessible Team, 2023). Improved web accessibility means more people can peruse, consume, and create content, leading to better exchanges and dissemination of information regardless of a user's abilities.

Key Takeaway

Web accessibility is a core component of effective digital services, ensuring that online content can be used across a wide range of abilities, devices, and contexts. By applying established accessibility principles and guidelines, libraries can improve usability and reduce barriers to digital resources.

Works Cited

ADA.gov. (2024, April 8). *Fact Sheet: New Rule on the Accessibility of Web Content and Mobile Apps Provided by State and Local Governments*. Retrieved from Americans with Disabilities Act U.S. Department of Justice Civil Rights Division: https://www.ada.gov/resources/2024-03-08-web-rule/#:~:text=Requirement:%20The%20Web%20Content%20Accessibility,the%20World%20Wide%20Web%20Consortium.

Alghamdi, A. M., Aljedaani, W., Jalali, H., Ludi, S., & Eler, M. (2025). Understanding developer challenges and trends in web accessibility: a stack overflow analysis. *Universal Access in the Information Society*, 24, 1701–1717. https://doi.org/https://doi.org/10.1007/s10209-024-01174-3

AllAccessible Team. (2023, October 29). *Web Accessibility Statistics: The Impact of Disabilities on Web Use*. Retrieved from AllAccessible: https://www.allaccessible.org/blog/web-accessibility-statistics-the-impact-of-disabilities-on-web-use

Bureau of Internet Accessibility. (2019, August 22). *Is There a Legal Requirement to Implement WCAG?* Retrieved from Bureau of Internet Accessibility: https://www.boia.org/blog/is-there-a-legal-requirement-to-implement-wcag#:~:text=The%20Web%20Content%20Accessibility%20Guidelines,do%20have%20to%20be%20accessible.

Georgakas, D. (2023). *A11Y Unraveled: Become a Web Accessibility Ninja*. APress. https://doi.org/https://doi.org/10.1007/978-1-4842-9085-9

Milbergs, K. (2023, August 11). *30 Key Web Accessibility Statistics*. Retrieved from Accessibly: https://accessiblyapp.com/blog/web-accessibility-statistics/

Petrosyan, A. (2025, September 2). *Worldwide internet user penetration from 2014 to July 2025*. Retrieved from Statista: https://www.statista.com/statistics/325706/global-internet-user-

penetration/#:~:text=Global%20internet%20user%20penetration%202014%2D2025&text=As%20of%20July%202025%2C%2068.7,worldwide%20were%20around%205.65%20billion.

W3C. (2025, May 6). *Web Content Accessibility Guidelines (WCAG) 2.1*. Retrieved from World Wide Web Consortium (W3C): https://www.w3.org/TR/WCAG21/

WAI. (2023, November 29). *WCAG 2 at a Glance*. Retrieved from W3C Web Accessibility Initiative (WAI): https://www.w3.org/WAI/standards-guidelines/wcag/glance/

WAI. (2024, March 14). *Essential Components of Web Accessibility*. Retrieved from W3C Web Accessibility Initiative (WAI): https://www.w3.org/WAI/fundamentals/components/

WAI. (2025, April 3). *Accessibility Fundamentals Overview*. Retrieved from W3C Web Accessibility Initiative (WAI): https://www.w3.org/WAI/fundamentals/

WAI. (2025, July 15). *Introduction to Understanding WCAG 2.2*. Retrieved from W3C Web Accessibility Initiative (WAI): https://www.w3.org/WAI/WCAG22/Understanding/intro#understanding-the-four-principles-of-accessibility

Yesilada, Y., & Harper, S. (Eds.). (2019). *Web Accessibility: A Foundation for Research.*

5.

USER EXPERIENCE

Chelsea Bernholtz; Cait Duffy; and Christina Tharp

Learning Objectives

By the end of this chapter, learners will:

- identify technologies and design strategies that support equitable access;
- evaluate technologies based on accessibility, sustainability, and effectiveness; and
- apply empathy-driven principles to ensure inclusive, culturally responsive design.

User Experience (UX): UX "is... considering spaces, services, and processes from the end user's point of view" (Bignoli and Stara, 2021) and "everything that touches upon your experience..." (Nielsen Norman Group, 2016).

Accessibility as the Foundation

"Accessibility is building for people in all the ways they come to libraries. Whether a person identifies as disabled or not, everyone can have a disabling experience" (Drayer, 2023).

Accessibility is not just technical, it's a core design principle. Libraries can start by applying accessibility audits for websites and digital catalogs. Designing with accessibility in mind means:

- adding alt text and descriptive metadata;
- ensuring high-contrast, readable text; and
- testing usability with diverse patrons, including patrons with disabilities.

Building accessibility into every design decision naturally supports equitable technology choices.

Choosing Appropriate Technologies

"Meet people where they are, not where you think they should be. Listen to what people really need" (Bignoli and Stara, 2021).

Appropriate technology depends on user needs and context. Practical steps include:

- conducting surveys or focus groups to understand priorities;
- reviewing analytics to see how patrons search and access resources; and
- piloting tools in small steps before fully integrating.

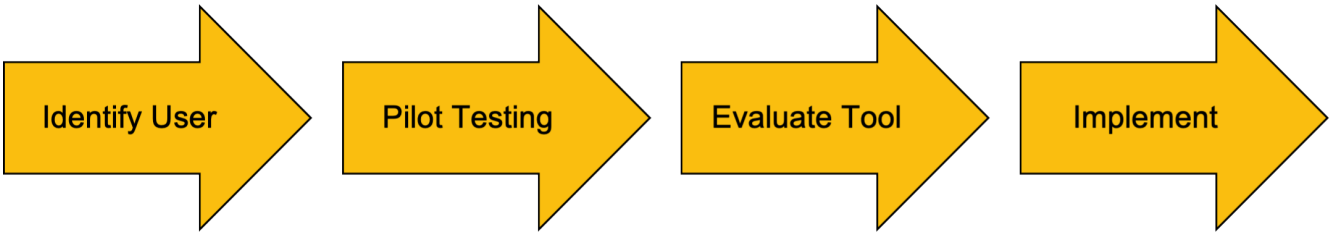


Figure 1. Before implementing technologies, librarians must first identify users, pilot test, and evaluate tools.

Selecting the right technology isn't just about features; it's about how it helps *all* patrons access and use library services. This helps librarians evaluate tools for their audiences.

Empathy and Ethical Design

"From a user-centered perspective, the new inner focus on users is both legitimate and beneficial to their needs, preferences, and growth potential in lifelong holistic information activities" (Nahl, 2017, p. 4824).



Empathy-based UX is both ethical and actionable. Libraries can integrate empathy by:

- hosting co-design sessions with users;
- providing staff training in cultural competence and accessibility; and
- collecting feedback and adapting services.

These strategies ensure technology serves patrons while addressing cultural needs. Involving diverse community members in usability testing builds culturally responsive systems, trust, and engagement.

Evidence, Evaluation, and Sustainability

"You can put these ideas into practice incrementally, starting with tiny changes and building up to larger, system-wide innovations" (Bignoli and Stara, 2021).

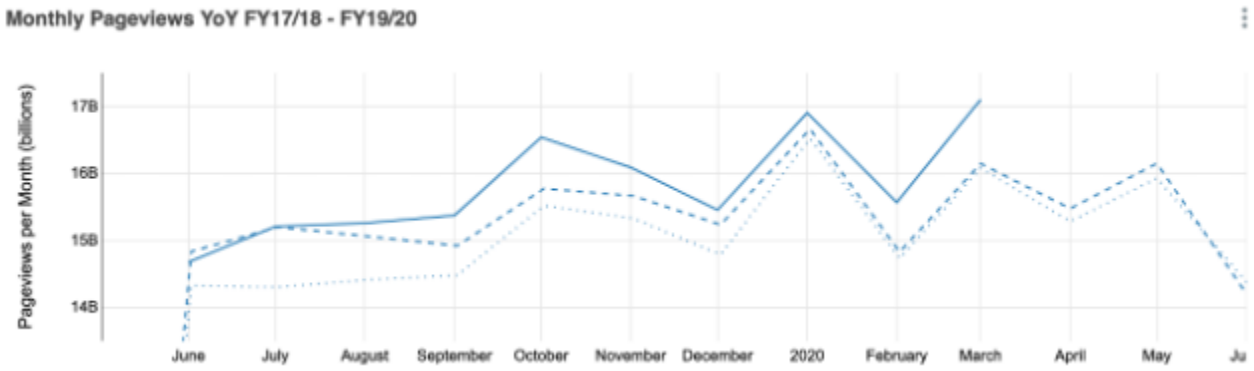


Figure 2. Chart depicting monthly pageviews

Practical evaluation includes:

- using analytics dashboards to track engagement;
- testing new features with pilot groups; and
- using results to adjust services.

By tracking results and testing new ideas on a small scale, libraries can see what works, reduce risk, and make sure their technology truly benefits patrons.

Key Takeaway

Designing library user experiences is about practical empathy, accessibility, and evidence. By applying targeted tools, involving users, and evaluating outcomes, libraries create systems that meet diverse needs.

Works Cited

Bignoli, C., & Stara, L. (2021, March 1). Users at the Center of Everything. *American Libraries Magazine*. https://americanlibrariesmagazine.org/2021/03/01/user-experience-center-of-everything/

Drayer, A. (2023, April 3). Digital Accessibility in Libraries: A Primer. *Choice 360*. https://www.choice360.org/libtech-insight/digital-accessibility-in-libraries-a-primer/

Nahl, D. (2017). User-Centered Revolution: 1970–1995 [ELIS Classic]. In J. D. McDonald & M. Levine-Clark (Eds.), *Encyclopedia of Library and Information Science* (4th ed., pp. 4812–4846). CRC Press. https://doi.org/10.1081/E-ELIS4-120044792

Nielsen Norman Group. [NNgroup]. (2016, July 2). *Don Norman: The term "UX"* [Video] YouTube. https://www.youtube.com/watch?v=9BdtGjoIN4E

6.

ONLINE REFERENCE INTERVIEWS

Lacey Johnson; Rachel Maly; and Sharee Snelson

Learning Objective

The intention of this chapter is to:

- define what an online reference interview is;
- learn about technologies that support access to online reference interviews;
- understand cultural and ethical considerations;
- understand guidelines and best practices of online reference interviews; and
- learn about emerging technology.

Libraries are becoming increasingly more digital. As the shift to a more digital space continues, the online reference interview is becoming critical for connecting users with information. The online reference interview presents different challenges than an in-person reference interview; use of technology, communication barriers, cultural and ethical considerations all need to be taken into account. When conducting an online reference interview the effective use of technology on both the librarian's side and the user can influence how one interacts. The interactions of the online reference interview must also be sensitive to ethical and cultural ways that users interpret and seek the information. Librarians must ensure that the tools needed to support digital reference services remain accessible and sustainable. This chapter will define online reference interview practices and challenges.

A student is in class working on a research project and sends an email to the librarian. The librarian receives the email and responds during her planning period with questions about their specific research goals. The student responds later in the day. The librarian can then provide better guidance and resources for the student. Is this an ideal scenario of an asynchronous online reference? What could be done better? How might the student's or librarian's background affect the reference interview?

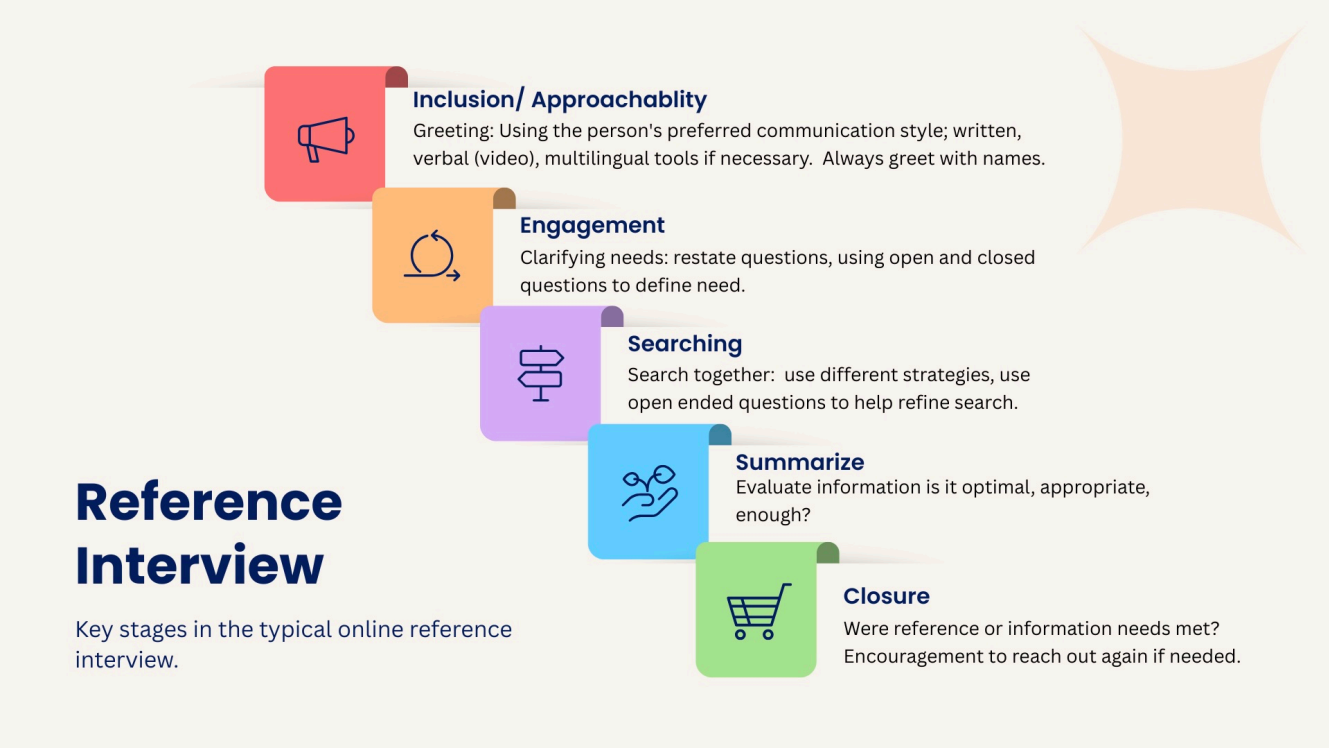


Figure 1. Infographic depicting the key stages in a typical online reference interview

Technologies

There are many different communication tools, both synchronous and asynchronous that allow flexibility and access. Synchronous tools such as library chat platforms (e.g., LibAnswers, LibChat, LiveHelpNow), video conferencing tools (e.g., Zoom, Google Meet), and screen sharing software (e.g., Zoom Share Screen, CoBrowse) provide real time––more like in-person––interviews. These can provide better results and are especially helpful for more detailed or unique search questions. Asynchronous communications tools such as email, online reference forms embedded within the libraries website, or ticketing systems allow for more flexibility across schedules and are generally accessible for most patrons.

Cultural and Ethical Considerations

When you begin planning, then conducting, and reviewing your online reference interview, you should be considerate of your interviewee's cultural and ethical views. It is important that before your interview takes place, you inform your interviewee of their informed consent, be transparent about the nature of the interview, and how you will keep their privacy during the interview. You should also be aware of your interviewee's time zone differences. Since you are conducting the interview, you should be willing to meet with your interviewee at a time that works best for them even if it means it's the middle of the night for you.

So, what is informed consent? Informed consent means that as the interviewer, you are to explicitly explain the nature of the interview and how it will be used. Some questions to consider are:

- Will it be shortened into an essay? If so, remember that quotes can often be misconstrued if taken out of context.
- Will the interview be placed online for others to see?
- Does the interviewee understand what the interview is about? How long will it take? Be transparent!

Once the interviewee is completely assured about the intent of the interview, they can give their verbal or written consent to conduct the interview.

Emerging Technology

As online reference interviews grow in popularity, technology grows, as well, to expand capabilities for both the interviewer and the interviewee. One of the ways that technology is growing at the fastest rate is through the use of artificial intelligence (AI). Through the use of AI, various tools are available for all parties involved in the interview. If the interview is going to be posted online, the interviewer may choose to have a background behind the involved parties so it looks like both parties are in the same room. Improved video quality is another way that technology is improving.

Key Takeaway

Online reference interviews extend traditional reference practices into digital environments and require careful attention to communication, technology, and user context. Effective online reference work depends on clear interaction, ethical awareness, and thoughtful selection of synchronous and asynchronous tools.

Work Cited

Lawton Henry, S. (2005, February). *Introduction to Web Accessibility*. Web Accessibility Initiative (WAI). Retrieved March 2, 2026, from https://www.w3.org/WAI/fundamentals/accessibility-intro/

7.

ONLINE SEARCH TECHNIQUES

Wendy Burkett; Molly Clearman; and Zach Lively

Learning Objectives

- Locate resources that describe advanced search techniques
- Apply advanced search techniques for information retrieval

How Database Searches Work

Patrons looking for specific research must navigate various search engines and databases to find appropriate material. Proper use of these search tools requires understanding of how databases and search engines work, what metadata is, and how to use advanced search tools.

Basic Search Functions

Basic searches require the search of keywords that narrow down relevant resources. Many engines and databases use what is called a "discovery layer" which consolidates resources into an all-in-one search box (Evans, 2021). Discovery layers were adopted by many libraries after the rise of search engines like Google. Unlike databases, Google is not curated by librarians. Google's search functions are more limited, particularly when using metadata, which will be discussed in the next section. The PDF linked below provides some tips for searching more efficiently within Google.

Google Search Tips

Advanced Searching

Databases are organized using metadata that breaks sources into various fields. Librarians assign fields to materials to organize their data and allow users to find the most relevant and timely information (Massachusetts Institute of Technology, n.d.).

Common fields:

- Title
- Author
- Subject
- Date
- Publication
- File type
- User rights

Searching for a database using keywords can yield many irrelevant sources. **Boolean searching** helps users find specific information and eliminates superfluous results.

Operators

Operators are additional words that assist in expanding or narrowing key words within searches. There are three primary operators:

- AND = in addition to
- OR = this or that
- NOT = exclude from search

Many advanced searches allow additional search boxes with these operators (see Figure 1). Operating words can also be typed within the search box, but they must be capitalized to avoid being filtered out as "stop words" or words that databases ignore in order to generate narrower searches.

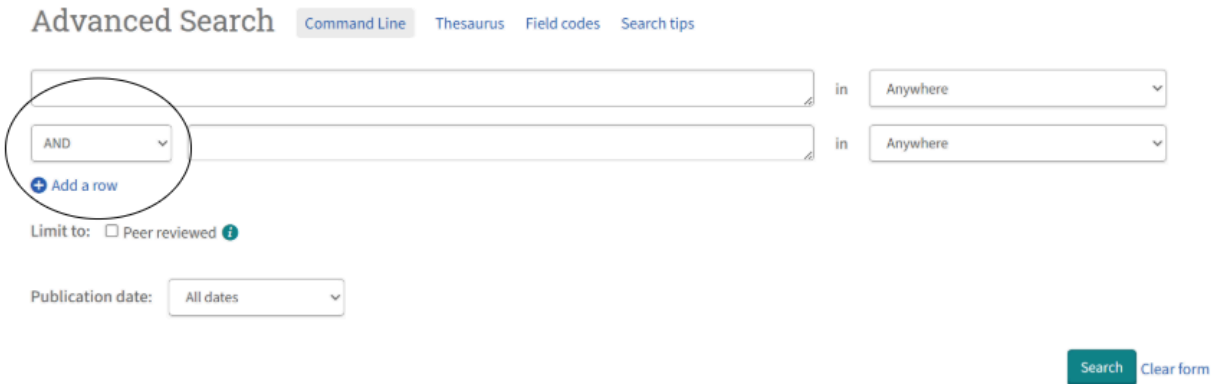


Figure 1. What the Boolean Operator "AND" looks like in a library database's advanced search.

These operators can be used together to comb databases for the most relevant sources. For example, a patron may be looking for information regarding contemporary literature. If they wanted to widen their search, they could use OR:

- Contemporary literature OR 21st century literature

AND can further narrow their search:

- Contemporary literature OR 21st century literature AND novels

These operators can be compounded into hyper-specific searches that selectively include and/or exclude terms. A more specific search may be:

- Contemporary literature OR 21st century literature AND novels NOT poetry

Quotations

Quotations are another tool that require a search to include each keyword with the same sequence. Searching "contemporary American novels" will only display sources that feature that sequence of words.

Truncation and Wild Cards

Symbols can be used to truncate words and enable the search to include alternate conjugations. The symbol varies by database, but the most common symbols are *, !, ?, or #.

- Example: Typing Limit* allows the results to include limit, limits, limited, limitless, etc.

Wild cards also account for alternate forms or spelling within sources.

- Example: Wom?n now enables the database to search for "women" or "woman"

The PDF linked below provides a simple breakdown of Boolean search techniques patrons may employ to aid their research.

Boolean Searching Basics

Key Takeaway

Understanding how search engines and databases organize information enables librarians to guide users toward more relevant results. By applying and explaining advanced search techniques, librarians support more efficient and informed research practices.

Works Cited

Columbia University Libraries. (2019). *Research guides: Database searching guide: Home*. Columbia University Libraries. https://guides.library.columbia.edu/databasesearching/home

Evans, G. (2021, December 20). *Good question! What is a discovery layer?* Ohio Technology Consortium. https://www.oh-tech.org/blog/good_question_what_discovery_layer

Massachusetts Institute of Technology. (n.d.). *Libguides: Database Search Tips: Overview*. Research guides & expert librarians. https://libguides.mit.edu/c.php?g=175963&p=1160724

Mr. Library. (2022, January 26). *Advanced searching (Boolean searching) in library databases*. [Video]. YouTube. https://www.youtube.com/watch?v=AUXBadzHpKQ

8.

ONLINE REFERENCE RESOURCES

Anesa Hajdarevic and Kolin Jasper

Learning Objective

By the end of this chapter, readers should be able to enumerate different common types of reference resources and be able to locate them

A reference resource provides a person with a summary of a topic and links for further research. Online Reference Resources are the same, but available through digital portals, databases, and other online websites. A patron with their own phone, computer, tablet or one of the libraries can access these resources. One of the easiest online reference tools is the internet, which can be used by simply looking up topics. Besides the internet, other online reference resources include bibliographical, periodical, ready references, educational, health, law, government, and business resources. This chapter's content covers online reference resources and their use-cases.

A bibliographical resource or bibliography is a list of materials that can be sorted by author, title, and subject. Each item within a bibliography records pertinent information about the source like publisher, publication date, etc. Periodical reference resources include newspapers, magazines, and journals. A good reference tool to find them is the Directory of Open Access Journals or the Free Library. Ready reference tools are the quick and easy references or links for websites that are on hand to use. A general example of this would be a website with a quick or ready reference section about their specific topic or further research if the reader is interested. A bibliography can sometimes act like a ready reference for a specific topic.

As libraries continue to bridge communities to information and technology, other specialized online reference tools expand the reach and access of knowledge. In education, OpenStax gives librarians, teachers, and students opportunities to explore free digital textbooks and supplemental tools to enhance their learning experiences, while also being simple to navigate, customizable, and easy to integrate into schools. Librarians can evaluate its effectiveness through improved digital literacy, resource sharing, and cost savings across their learning communities.

In the health field, access to reliable information can be considered life changing, and librarians play an essential role in guiding users toward trustworthy sources. PubMed provides free access to over 39 million

citations of biomedical literature, including journal articles and clinical studies, and allows users to easily search by keyword, author, or medical subject heading to locate relevant research for their medical queries. Usage of these resources is beneficial for librarians as we can promote health literacy, accessibility, and enhance informed decision-making across communities.

Libraries are often the first stop for patrons seeking law and business resources. Two good examples are Modern Courts offering clear, accessible summaries of court structures and procedures, geared towards librarians to use when helping the public, and Business Dynamic Statistics (BDS) providing free, up-to-date data on job creation, firm births, and shutdowns through the BDS Explorer, supporting research and planning.

Government reference resources are materials published by the government's expenses or required by law. These documents are often separated by branches of government, departments, and other distinctions. A good example of a government reference resource is the Federal Register which provides regulations, rules, and notices from government agencies and departments.

Key Takeaway

Online reference resources provide quick access to reliable information across many subject areas and use cases. Familiarity with common resource types allows librarians to connect patrons with appropriate entry points for research and learning.

Resources

- Directory of Open Access Journals
- The Free Library by Farlex
- OpenStax
- PubMed
- The Fund for Modern Courts: A Guide to Legal Information and Resources
- United States Census Bureau: Business Dynamics Statistics (BDS)
- Federal Register

9.

USER TECHNOLOGY MANAGEMENT

Olivia Bishop; Kris Rausch; and Ana Vazquez Pryor

Learning Objectives

- Describe the importance of public computing policies
- Identify and explain methods for ensuring patron data privacy
- Describe effective ways to incorporate information technology (IT) into strategic planning
- Locate and evaluate credible sources

Policy & Librarians' Expectations

Librarians are responsible for providing safe, equitable access to technology for all. When creating policies regarding technology management, librarians should consider:

- **Timing:** Consider how often you want this plan to be reviewed and updated. This can be as soon as annually, or every three to five years, depending on what you see fit.
- **Accessibility:** Consider access to both information and physical space. It's important to create guides that help patrons navigate applications and optimize them based on their personal preferences.
- **Support:** Both patrons and library staff need support. Management policies can educate staff through workshops, training, and additional supports that can aid librarians in utilizing and educating patrons on technology.

Monitoring & Privacy

Librarians are responsible for upholding the American Library Association (ALA) Library Bill of Rights, the library's policies, and the protection of patron privacy. According to the *Access to Digital Resources and Services: An Interpretation of the Library Bill of Rights*, publicly funded libraries must provide access to constitutionally protected information (American Library Association [ALA], 2025). Libraries may use filtering programs to limit access to illegal materials, although these limitations should be used sparingly. Filtering internet access can restrict users if the content they are searching for contains sensitive information (Edlund, 2020). Clear,

publicly available policies leave little room for misunderstanding by library staff and patrons, providing the library with a foundation to enforce their guidelines.

User privacy should remain a grounding principle for managing technology systems in the library. The right to free inquiry without judgment or scrutiny is protected by the First Amendment and should be protected by the library and its staff, including utilizing malware tools to protect patron information and informing patrons about why and how their information is being stored (ALA, 2025).

Planning

Part of what librarians are responsible for is the planning that goes into technology management. There is a fine line between being a repository and an active facilitator of information (ALA, 2025). As technology evolves, libraries must adapt to emerging needs. For example, there has been a rise in technologies using artificial intelligence that can produce immediate results, which may or may not be accurate. Thus, the library should design an AI policy and educate users about its appropriate use.

Library patrons have a wide array of technological needs. Some users may need step-by-step instructions on how to log in to their emails. Others might be doing niche research and simply need to be pointed to an appropriate database. Policies should encompass a wide audience range, and that can vary based on what type of library it is (and the resources they have).

Credible Sources

When developing public computing policies, librarians should locate and evaluate credible sources for guidance. Sources should include ALA guidelines, state regulations, and professional best practices.

Key Takeaway

User technology management relies on clear policies, attention to privacy, and ongoing planning. Thoughtful management practices help libraries balance access, security, and sustainability while supporting consistent public service.

Works Cited

American Library Association. (2025, May 9). *Access to Digital Resources and Services: An Interpretation of the Library Bill of Rights*. https://www.ala.org/advocacy/intfreedom/librarybill/interpretations/digital

American Library Association. (2019, June 24). *Guidelines for Library Policies*. https://www.ala.org/advocacy/intfreedom/librarypolicyguidelines

American Library Association. (2020, January 26). *Library Privacy Guidelines for Public Access Computers and Networks*. https://www.ala.org/advocacy/privacy/guidelines/public-access-computer

American Library Association. (2019, June 29). *Privacy: An Interpretation of the Library Bill of Rights*. https://www.ala.org/advocacy/intfreedom/librarybill/interpretations/privacy

Edlund, H. (2020). An analysis of American public libraries' policies on patron use of Internet pornography. *Open Information Science. Vol. 4*(1), 58-74. https://doi.org/10.1515/opis-2020-0005

PART II

THE DIGITAL DIVIDE

10.

INTRODUCTION TO PART II: THE DIGITAL DIVIDE

Part II builds on the foundational concepts introduced in Part I by examining how differences in access, skills, infrastructure, and resources affect patrons' use of technology in library contexts. While Part I focuses on shared practices and services across libraries, the chapters in Part II take a more applied and contextual approach.

Each chapter in this section considers how technology use may vary across specific populations or settings, highlighting practical challenges that librarians are likely to encounter in public-facing work. As a result, these chapters may engage more directly with social, economic, geographic, or institutional factors that shape patrons' interactions with digital tools and services.

Readers may find that chapters in Part II vary in depth, structure, and emphasis. This variation reflects the complexity of the topics addressed and the different bodies of research and professional literature that inform them. Together, the chapters provide a set of perspectives that complement the service frameworks outlined in Part I and support a more comprehensive understanding of how libraries respond to differences in technology access and use.

11.

DIGITAL DIVIDE AND AGE

Sophia Craig; Natalie Goble; Caroline Oberg; Abby Wirtz; and Dani Youngberg

Learning Objective

By the end of this chapter, readers will understand how age contributes to the digital divide, specifically relating to the access, digital literacy, and attitudes of different generations toward technology. They will also be made aware of potential strategies to help bridge the generational digital divide in library settings

Introduction

The range of digital literacy and skills across generations is an integral factor in the digital divide. Although children today are often introduced to technology like iPads, streaming services, and the internet at a young age, older adults, on the other hand, have lived most of their lives without access to today's technology. Many parents today may contribute to early technology adoption in the home, while schools do so in the classroom, some as soon as kindergarten. Unfortunately, because older generations were not exposed to it at a younger age, they are more hesitant to learn how to access and use technology. To them, the digital presence in everyday life is a discouraging but growing force that prevents them from participating in democracy, managing finances, and even connecting with their family and friends (Lamphere, 2019).

When looking at the role of age in the digital divide, most research shows that it primarily impacts older generations. This gap between older adults and younger generations shows that the older population have fallen behind with technology for a handful of reasons. The Centre for Ageing Better found that there were five major contributors that impacted older adults' use of technology: self-confidence, fear, physical issues, cultural differences, and lack of social networking (Wilson, 2020). Older users may have mental and/or physical disabilities that factor into their struggle with digital advances. While people who struggle with digital literacy are often labeled as having negative attitudes towards technology, it is more often based on nervousness and not knowing where to start.

The Myth of the Digital Native

Younger generations who have grown up with advanced forms of technology are more likely to understand

how to use them. Many assume young people's early exposure to screens is a path to generating and sharing more complex media content, sometimes referring to them as "digital natives." However, this does not mean that all younger users have access to or know how to use this technology. Some researchers are actively trying to dispel the myth of the digital native because even among young people, it is "possible to find all kinds of situations, profiles and experiences of digital technology use" (Scolari, 2019). Similarly, not all baby boomer or Gen X patrons live with disabilities that affect their use of technology, and some may live in households where they are exposed to digital media often. Research shows how access to everyday technology among older adults is increasing every year (Faverio, 2022).

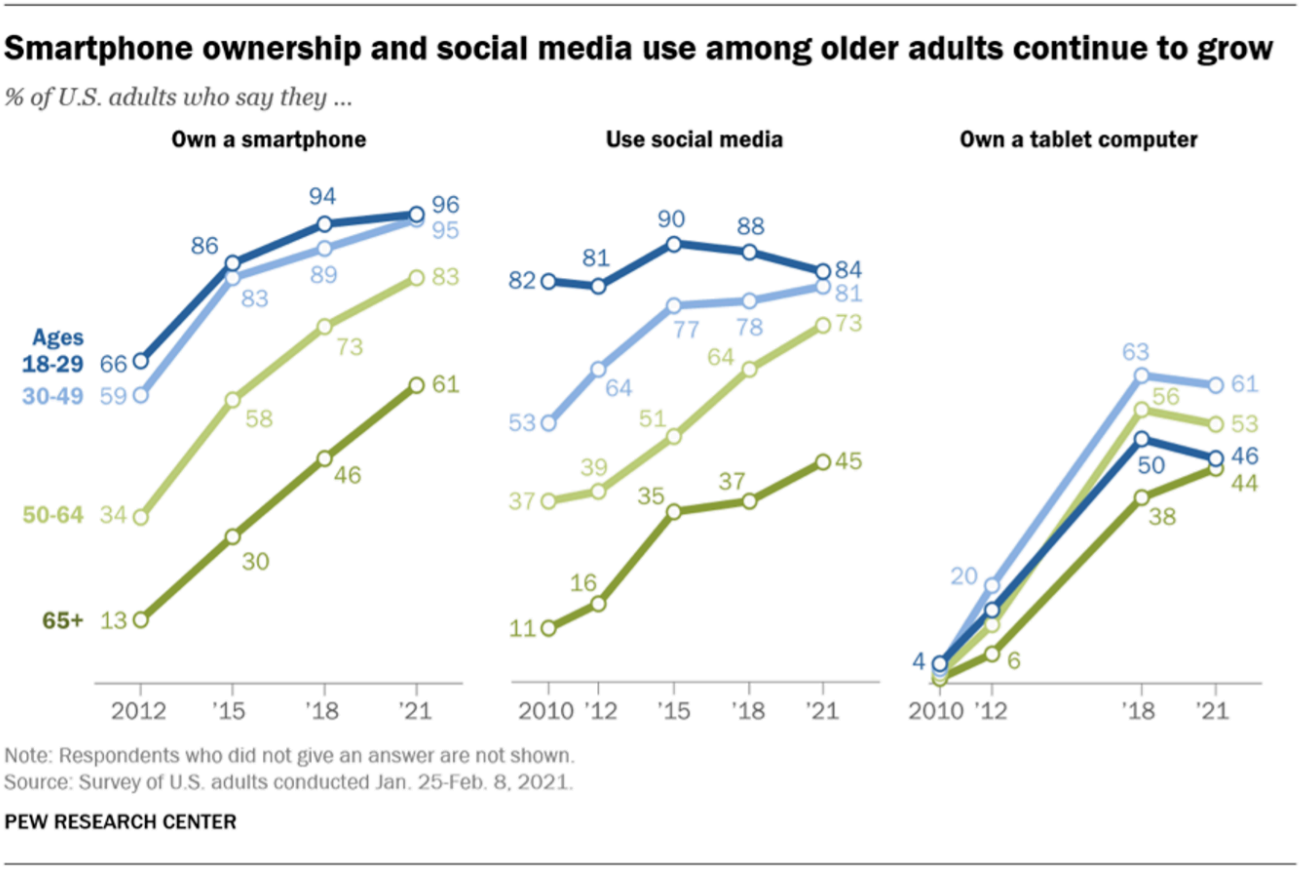


Figure 1. "Smartphone ownership and social media use among older adults continue to grow." Pew Research Center, Washington, D.C. (2022, January 13).

How Libraries and Schools Are Bridging the Divide

The divide between age groups is narrowing, but the divide still exists. Libraries and schools are fighting to mitigate the factor of age in the digital divide by providing resources to help those who have trouble understanding, using, and evaluating technology and digital information. Young people need to learn how to be able to use and successfully navigate the changing digital landscape, as so much of everyday life now has to be done on the computer, including education, job applications, taxes, and more. Because of this, students must

have the basic skills to be able to adapt and operate platforms they've never been on. Schools are continuing to implement updated technology standards to ensure students are receiving the most current education on digital information possible, as well as beginning to utilize external resources, like Code.org, which comes to K-12 classrooms to help increase access and encourage computer science in classrooms (Rogers, 2016).

Many libraries provide programs and classes to help bridge the digital divide, from basic skills like keyboarding and using a mouse to advanced skills like website design and coding. The Brooklyn Public Library developed a curriculum called 'Never Too Late to Learn' that offers free technology classes and assistance to patrons. This resource caters specifically to those over the age of fifty by "paying attention to physical limitations in vision, hearing, and mobility that could prove frustrating in a learning environment"(Witteveen, 2017). One of its most popular programs is the virtual bowling league called 'Library Lanes', which uses the Xbox One gaming system. For homebound patrons, they also provide tech programs online and via teleconference.

More and more libraries are also starting to build digital media labs where patrons can learn creative technology like Adobe Creative Suite, Cricut, and music production software. Additionally, some libraries offer one-on-one technology help sessions where a user can bring in their device and get assistance from a librarian or IT professional (Lamphere, 2019). While it is true that there is a large population of people in older generations who are negatively impacted by the digital divide, people in younger generations aren't immune from falling behind either, as another big factor in the digital divide is people's socioeconomic status, which will be discussed in a later chapter.

For more information and further reading related to age and the digital divide, see the resource list below:

1. Computer Classes, from the DC Public Library, example of library technology classes
2. Gen Z and Digital Inclusion, report from the Digital Future Society
3. Social media communication and shifting attitudes toward social change across generations: a comparative study of Gen Z and older cohorts

Key Takeaway

Age can influence how individuals access, use, and perceive technology, but generational differences are varied and not fixed. Libraries address age-related gaps by offering instruction and services that accommodate different experience levels and learning preferences.

Works Cited

Faverio, M. (2022, January 13). Share of those 65 and older who are tech users has grown in the past decade. *Pew Research Center*. https://www.pewresearch.org/short-reads/2022/01/13/share-of-those-65-and-older-who-are-tech-users-has-grown-in-the-past-decade/

Lamphere, C. (2019). Libraries, Seniors, and the Challenge of Technological Education and Outreach. *Online Searcher, 43*(1), 51–54.

Rogers, S. E. (2016). Bridging the 21st Century Digital Divide. *TechTrends, 60*(3), 197–199. https://doi.org/10.1007/s11528-016-0057-0

Scolari, C. A. (2019). Beyond the myth of the "digital native": Adolescents, collaborative cultures and transmedia skills. *Nordic Journal of Digital Literacy, 14*(3–4), 164–174. https://doi.org/10.18261/issn.1891-943x-2019-03-04-06

Wilson, G. (2020, June 2). How the Digital Divide Affects Older Adults' Use of Technology During COVID-19. *Centre for Ageing Better*. https://ageing-better.org.uk/blogs/how-digital-divide-affects-older-adults-use-technology-during-covid-19

Witteveen, A. (2017). Senior Partners: Innovation. *Library Journal*. www.libraryjournal.com/story/senior-partners-innovation.

12.

DIGITAL DIVIDE AND DISABILITY

Sera-Ann Hargrove; Hannah Highsmith; Page Sutton; and Erin Traicoff

Learning Objective

By the end of this chapter, readers will understand how people with disabilities experience the digital divide, the barriers that limit access to digital participation, and how libraries can foster digital equity through accessibility initiatives, technology, and advocacy. Readers will also be able to evaluate the social and structural factors that contribute to digital inequity and consider the evolving role of libraries as agents of empowerment in the information society.

Explore

Pew Research Center chart: "Americans with disabilities less likely than those without to own some digital devices"

Chart from Pew Research Center; details follow in caption.

Figure 1. "Americans with a disability are less likely than those without one to have traditional computer, smartphone." Pew Research Center, Washington, D.C. (2021, September 10).

Understanding Digital Accessibility: Disability as a Cause of Digital Exclusion (Video)

One or more interactive elements has been excluded from this version of the text. You can view them online here: https://pressbooks.uiowa.edu/pust/?p=26#oembed-1

Accessibility in Technology and Design: When we design for

disability, we all benefit (Video)



One or more interactive elements has been excluded from this version of the text. You can view them online here: https://pressbooks.uiowa.edu/pust/?p=26#oembed-2

Libraries and Accessibility: 4 Ways to Design a Disability-Friendly Future (Video)



One or more interactive elements has been excluded from this version of the text. You can view them online here: https://pressbooks.uiowa.edu/pust/?p=26#oembed-3

Examined Life – Judith Butler & Sunaura Taylor (Video)



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A family visit to the library (Video)



One or more interactive elements has been excluded from this version of the text. You can view them online here: https://pressbooks.uiowa.edu/pust/?p=26#oembed-5

Resources

- AChecks Web Accessibility Checker
- W3C Accessibility Principles

- W3C Introduction to Web Accessibility

Introduction: *The Problem*

> "People with disabilities are perhaps the single segment of society with the most to gain from new technologies of the electronic age. Yet they have among the lowest rates of use of these technologies" (Scanlan, 2022, p. 726).

Technology, often celebrated as the great equalizer, paradoxically reinforces inequities when access is uneven. As society becomes increasingly digital, participation in everyday life—securing employment, obtaining education, engaging in civic discourse, and connecting socially—relies on reliable internet access and functional, usable devices. For many people with disabilities, this reliance presents significant barriers.

According to the Centers for Disease Control and Prevention (2024), disability refers to "any condition of the body or mind (impairment) that makes it more difficult for the person with the condition to do certain activities and interact with the world around them." This broad definition captures physical, sensory, cognitive, and mental disabilities, each of which intersects with technology differently. Yet, regardless of type, people with disabilities often experience compounded exclusion in digital contexts because technologies are not designed with their lived experiences in mind.

The result is a cycle of digital inequity: lack of access limits educational and employment opportunities, which constrains income, which in turn limits the ability to afford devices and connectivity. Over time, this cycle reproduces systemic poverty and social isolation. As more aspects of life such as applying for benefits, paying bills, or accessing telehealth move online, digital exclusion increasingly equates to civic invisibility.

Key Findings: Barriers to Access

Affordability

Economic barriers remain one of the most persistent obstacles to digital inclusion. Devices such as computers, smartphones, or tablets are costly, and high-speed internet remains financially out of reach for many individuals with disabilities who experience disproportionately high rates of unemployment or underemployment. The U.S. Department of Labor found that only 78.4% of households with a disabled adult had home internet, compared to 91.5% of households without (Fong & Swick, 2022). For people who rely on disability benefits, which often fall below the poverty line, digital participation competes with necessities like food, rent, and healthcare.

Assistive Technology and Accessibility

Even when internet access exists, technological accessibility does not. Assistive technologies like screen readers,

speech recognition software, refreshable braille displays, captioning tools, and adaptive keyboards remain prohibitively expensive, inconsistently supported, or incompatible with mainstream platforms (Chadwick, Wesson, & Fullwood, 2013). Furthermore, accessible design is frequently treated as an afterthought rather than a design principle. Without accessibility "baked in" from the beginning, websites and digital tools often require retrofitting that is partial, costly, and ineffective.

Design Exclusion

This lack of foresight represents a deeper ideological barrier: the tendency to imagine a "default user" who is able-bodied, neurotypical, and resourced. In this sense, the digital divide is not just technological—it is cultural. When developers and institutions fail to account for diverse bodies and minds, they inadvertently construct systems that exclude. Scholars like Meiselwitz, Wentz, and Lazar (2009) remind us that universal usability requires not only accessible code but an accessible mindset.

Statistics: Measuring the Gap

Quantitative data underscores the severity and persistence of the divide. In addition to the Department of Labor's findings, a Pew Research Center study found that Americans with disabilities are 19% less likely to own a computer and 16% less likely to own a smartphone than people without disabilities (Atske & Perrin, 2021).

These numbers are not abstract—they translate into lived limitations. For instance, individuals with mobility impairments may depend on online platforms for work or healthcare yet lack the adaptive hardware to participate fully. For people with visual impairments, poorly coded websites without alt text or screen reader compatibility become impenetrable. Meanwhile, deaf or hard-of-hearing users are excluded from video-based media when captions are unavailable.

The divide also intersects with other forms of marginalization. Research suggests that people with disabilities who are also members of racial or linguistic minority groups face compounded barriers, as systemic inequities amplify the effects of disability discrimination. Thus, digital exclusion cannot be disentangled from broader questions of social justice and equity.

Library Strategies for Bridging the Divide

Libraries, long called the "great equalizers" of information, now play a vital role in ensuring equitable access to technology for people with disabilities. They can bridge the digital divide through accessible tools, trained staff, inclusive programs, and community collaboration.

1. **Assistive Technology:** Provide devices and software such as screen readers, braille printers, and magnifiers. Equally important is promoting these resources so patrons know they are available (Chadwick, Wesson, & Fullwood, 2013).
2. **Staff Training:** Librarians should understand how technology works, with each staff member becoming familiar with at least one assistive tool. Programs like Certified Inclusion Ambassadors help make accessibility a shared team competency (Guder, 2014; Holland, 2024).
3. **Inclusive Programming:** Offer workshops and events that prioritize accessibility, such as sensory-friendly storytimes, coding classes, and digital literacy sessions for assistive technology users. These programs empower patrons beyond mere access.
4. **Community Partnerships:** Collaborate with disability organizations and advocacy groups to design services that meet real needs. For example, Joeten-Kiyu Public Library used grant funding to create a Sensory Corner with adaptive technology, illustrating how partnerships can drive meaningful inclusion (Marcotte, 2024).

Policy Implications

Bridging the disability digital divide cannot rely solely on institutional goodwill; it requires structural reform. Policy must evolve to reflect digital participation as a right, not a privilege.

1. **Accessibility Standards:** Stronger enforcement of the Americans with Disabilities Act (ADA), Section 508, and Web Content Accessibility Guidelines (WCAG) is essential. Compliance must be seen not as a checklist but as a moral baseline for digital citizenship.
2. **Universal Design as Policy:** Embedding accessibility into procurement, software development, and institutional planning ensures sustainability. Rather than "fixing" inaccessible technologies after launch, universal design anticipates human diversity from the start (Meiselwitz, Wentz, & Lazar, 2009).
3. **Sustainable Funding for Digital Equity:** Grants and federal programs that fund assistive technologies, digital literacy training, and broadband access in public institutions are critical. As Grimes and Porter (2024) argue, digital equity is a collective project that depends on sustained investment and policy alignment.
4. **Civic Framing of Digital Inclusion:** Ultimately, access for people with disabilities should be understood as a civil rights issue (Scanlan, 2022). Libraries, through their public trust and inclusive mission, are natural advocates for this framing. By positioning themselves as civic anchors, libraries can influence broader public policy and reframe accessibility as a cornerstone of democratic participation.

Conclusion

Technology's promise lies in its potential to connect, empower, and transform. Yet that promise remains incomplete when significant segments of the population (particularly people with disabilities) remain excluded from participation. The digital divide is not merely about internet connections or devices; it is about who is seen, who is heard, and who is counted.

Libraries, with their dual commitment to information and equity, embody the practical and ethical response to this challenge. By embracing assistive technologies, developing staff expertise, creating inclusive programming, and engaging in policy advocacy, libraries can transform the digital divide from a site of exclusion into a site of collective progress.

Digital equity is not a technical fix, but a societal commitment. In ensuring that people with disabilities can fully access, shape, and benefit from the digital world, libraries reaffirm their foundational promise: that knowledge, connection, and opportunity belong to everyone.

Works Cited

Atske, S., & Perrin, A. (2021, September 10). Americans with disabilities less likely than those without to own some digital devices. *Pew Research Center*. https://www.pewresearch.org/short-reads/2021/09/10/americans-with-disabilities-less-likely-than-those-without-to-own-some-digital-devices/

Bertot, J. C., & Jaeger, P. T. (2015, October 5). The ADA and inclusion in libraries. *American Libraries Magazine*. https://americanlibrariesmagazine.org/blogs/the-scoop/ada-inclusion-in-libraries/

Bowker, N., & Tuffin, K. (2003). Dicing with deception: People with disabilities' strategies for managing safety and identity online. *Journal of Computer-Mediated Communication, 8*(2). https://doi.org/10.1111/j.1083-6101.2003.tb00200.x

Centers for Disease Control and Prevention. (2024, April 3). *Disability and health overview*. https://www.cdc.gov/ncbddd/disabilityandhealth/disability.html

Chadwick, D., Wesson, C., & Fullwood, C. (2013). Internet access by people with intellectual disabilities: Inequalities and opportunities. *Future Internet, 5*(3), 376–397. https://doi.org/10.3390/fi5030376

Dobransky, K., & Hargittai, E. (2006). The disability divide in Internet access and use.

Information, Communication & Society, 9(3), 313–334. https://doi.org/10.1080/13691180600751298

Domingo, M. C. (2012). An overview of the Internet of Things for people with disabilities. *Journal of Network and Computer Applications, 35*(2), 584–596. https://doi.org/10.1016/j.jnca.2011.10.015

Ellis, K., & Kent, M. (2011). Disability and new media. *Routledge*. https://doi.org/10.4324/9780203835634

Fong, A., & Swick, S. (2022). Disability and the digital divide: Internet subscriptions, internet use and employment outcomes. *U.S. Department of Labor, Office of Disability Employment Policy*. https://www.dol.gov/sites/dolgov/files/ODEP/pdf/disability-digital-divide-brief.pdf

Grimes, N. D., & Porter, W. (2024). Closing the digital divide through digital equity: The role of libraries and librarians. *Public Library Quarterly, 43*(3), 307–338. https://doi.org/10.1080/01616846.2023.2251348

Guder, C. S. (2014). Making the right decisions about assistive technology in your library. In C. Booth (Ed.), *Making libraries accessible: Adaptive design and assistive technology* (7th ed., pp. 22–37). American Library Association.

Holland, A. (2024, March 1). Beyond ADA compliance. *American Libraries Magazine*. https://americanlibrariesmagazine.org/2024/03/01/beyond-ada-compliance/

Jaeger, P. T. (2012). Disability and the Internet: Confronting a digital divide. *Boulder: Lynne Rienner Publishers*.

Kleban, M., & Kaye, H. S. (2015). The use of online social networks and online communities among people with disabilities. *Disability and Health Journal, 8*(4), 419–425. https://doi.org/10.1016/j.dhjo.2015.03.001

Marcotte, A. (2024, November 1). Libraries transforming communities, one year later. *American Libraries Magazine*. https://americanlibrariesmagazine.org/2024/11/01/libraries-transforming-communities-one-year-later/

Meiselwitz, G., Wentz, B., & Lazar, J. (2009). Universal usability: Past, present, and future. *Foundations and Trends in Human–Computer Interaction, 3*(4), 213–333. https://doi.org/10.1561/1100000027

Scanlan, M. (2022). Reassessing the disability divide: Unequal access as the world is pushed online. *Universal Access in the Information Society, 21*(3), 725–735. https://doi.org/10.1007/s10209-021-00803-5

13.

DIGITAL DIVIDE AND IMMIGRANTS

Skyler Appler; Emilia F. Gaps; Charles Stuart Kline; Jenna Ross; and Laura Angelica Velazquez-Casillas

Learning Objective

Understand how the digital divide affects immigrant communities and identify strategies libraries can use to promote digital inclusion and equitable access to technology.

Immigration and Library Practices

For decades, public libraries have supported immigrants by upholding their commitment to improve quality of life for all patrons—regardless of origin or citizenship, as supported by the Library Bill of Rights (American Library Association, 2019). Immigration is not a monolith; language, culture, education, socioeconomic status, and accessibility all shape how the digital divide manifests within immigrant communities. From ESL courses to digital lending and community partnerships, libraries meet immigrants where they are and help bridge gaps in digital access (Igarashi, Koizumi, & Widdersheim, 2023).

Explore more: Webinar: Immigrants in Academic Libraries (1 hour)

Barriers to Digital Inclusion

Socioeconomic and Education Factors

Educational attainment varies widely among immigrants. Among immigrant adults age 25 and older, 24% lacked a high school degree (compared to 7% of their U.S.-born cohorts). But immigrants are just as likely to have a bachelor's degree or higher (Kramer & Passel, 2025). Education level often determines digital skill and access. Median weekly earnings for foreign-born workers are lower, restricting their ability to invest in current technology or broadband (U.S. Department of Labor, 2025).

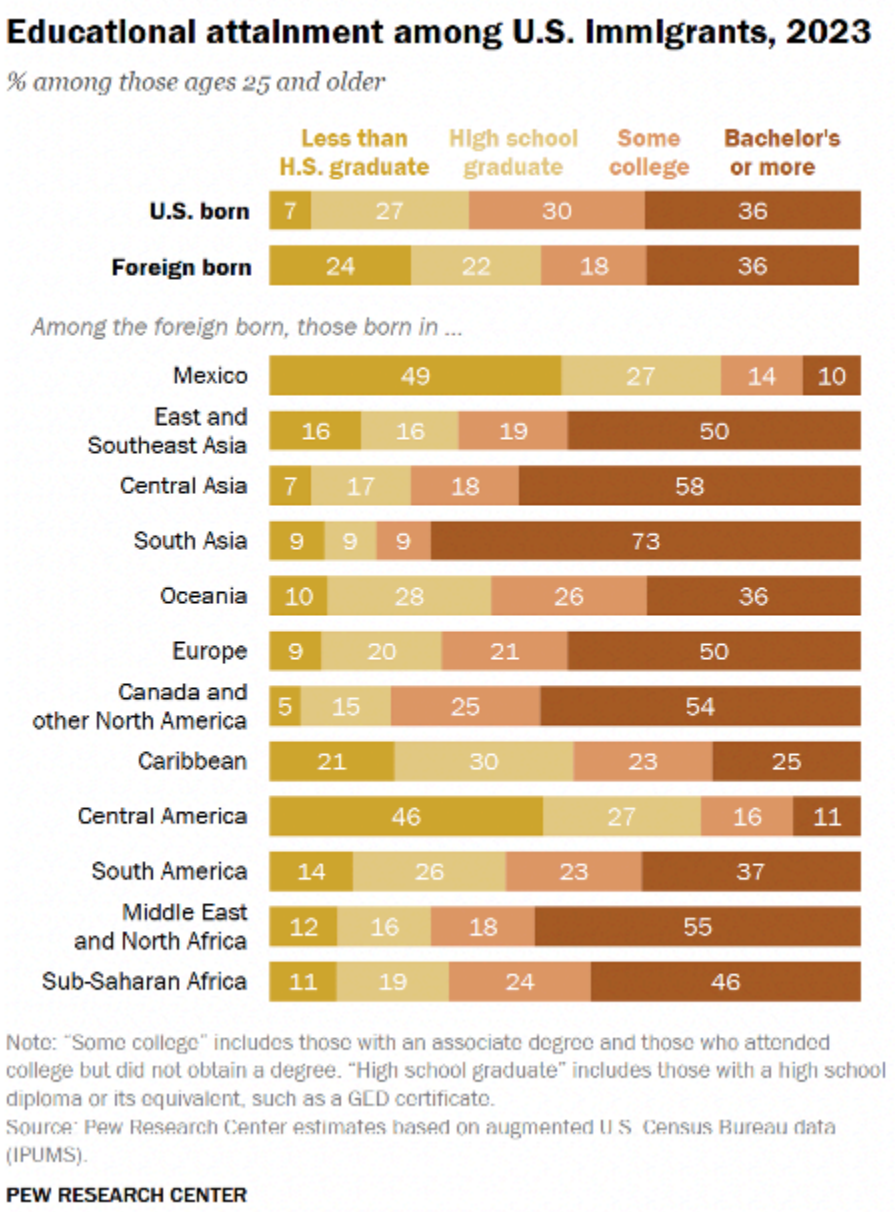


Figure 1. "Education attainment among U.S. Immigrants, 2023." Pew Research Center, Washington, D.C. (2025, August 21).

Only 31% of Hispanic households have broadband internet, and just 67% own a laptop, compared with 80% of white adults (Geraghty, Panettieri, Pedrosa, & Luna, 2021). Immigrants make up one-third of U.S. workers without digital skills (Cherewka, 2020). These gaps directly affect access to remote work, banking, and online services. Libraries can mitigate such disparities through free Wi-Fi, device lending, and digital literacy programs tailored to community needs.

Table 5. Median usual weekly earnings of full-time wage and salary workers for the foreign born and native born by selected characteristics, 2023-2024 annual averages

[Numbers in thousands]

Characteristic	2023					2024				
	Foreign born		Native born		Earnings of foreign born as percent of native born	Foreign born		Native born		Earnings of foreign born as percent of native born
	Number	Median weekly earnings	Number	Median weekly earnings		Number	Median weekly earnings	Number	Median weekly earnings	
Total, 16 years and over	22,675	$987	98,232	$1,140	86.6	23,319	$1,001	96,734	$1,190	84.1
Men	13,447	1,051	53,253	1,238	84.9	13,781	1,077	52,160	1,316	81.8
Women	9,228	899	44,978	1,025	87.7	9,538	911	44,573	1,074	84.8
AGE										
16 to 24 years	1,178	672	10,247	718	93.6	1,274	707	9,544	749	94.4
25 to 34 years	4,492	992	25,257	1,054	94.1	4,594	995	24,659	1,117	89.1
35 to 44 years	6,184	1,059	22,518	1,301	81.4	6,395	1,085	22,817	1,341	80.9
45 to 54 years	5,839	1,033	19,772	1,326	77.9	5,931	1,062	19,479	1,396	76.1
55 to 64 years	3,905	976	16,126	1,266	77.1	4,042	1,013	15,776	1,319	76.8
65 years and over	1,077	898	4,312	1,131	79.4	1,084	926	4,459	1,206	76.8
RACE AND HISPANIC OR LATINO ETHNICITY[1]										
White non-Hispanic or Latino	3,275	1,380	67,748	1,224	112.7	3,240	1,483	66,510	1,274	116.4
Black non-Hispanic or Latino	2,482	971	12,893	919	105.7	2,510	971	12,561	966	100.5
Asian non-Hispanic or Latino	5,904	1,503	2,489	1,464	102.7	6,003	1,532	2,404	1,541	99.4
Hispanic or Latino ethnicity	10,716	792	12,117	947	83.6	11,298	809	12,274	997	81.1
EDUCATIONAL ATTAINMENT										
Total, 25 years and over	21,497	1,012	87,984	1,206	83.9	22,046	1,027	87,190	1,259	81.6
Less than a high school diploma	3,694	692	2,353	748	92.5	3,813	717	2,287	789	90.9
High school graduates, no college[2]	5,298	809	21,029	919	88.0	5,538	821	20,594	962	85.3
Some college or associate degree	3,109	944	23,418	1,027	91.9	3,133	954	22,871	1,074	88.8
Bachelor's degree and higher[3]	9,397	1,637	41,185	1,602	102.2	9,562	1,738	41,437	1,679	103.5

[1] Data for race/ethnicity groups do not sum to totals because data are not presented for all races.
[2] Includes people with a high school diploma or equivalent.
[3] Includes people with bachelor's, master's, professional, and doctoral degrees.
NOTE: Updated population controls are introduced annually with the release of January data.

Figure 2. "Median usual weekly earnings of full-time wage and salary workers for the foreign born and native born by selected characteristics, 2023-2024 annual averages." U.S. Department of Labor, Bureau of Labor Statistics. (2025, May 20).

📖 Explore More: New Americans Initiative – List of Resources

Language, Culture, and the Library as a Third Space

Almost half of immigrants are proficient in English (Kramer & Passel, 2025), but limited English proficiency can deepen the digital divide. Shame or fear of appearing digitally illiterate may discourage patrons from asking for help. As Jessamyn West notes, a librarian's attitude can be pivotal in creating trust and belonging. Libraries serve as third spaces—neutral, community-driven environments where multigenerational immigrant families learn together. Younger generations often teach elders digital skills learned in school (Salazar & Sanchéz,

2011). Multigenerational programming, childcare support, and bilingual staff all enhance participation and confidence.

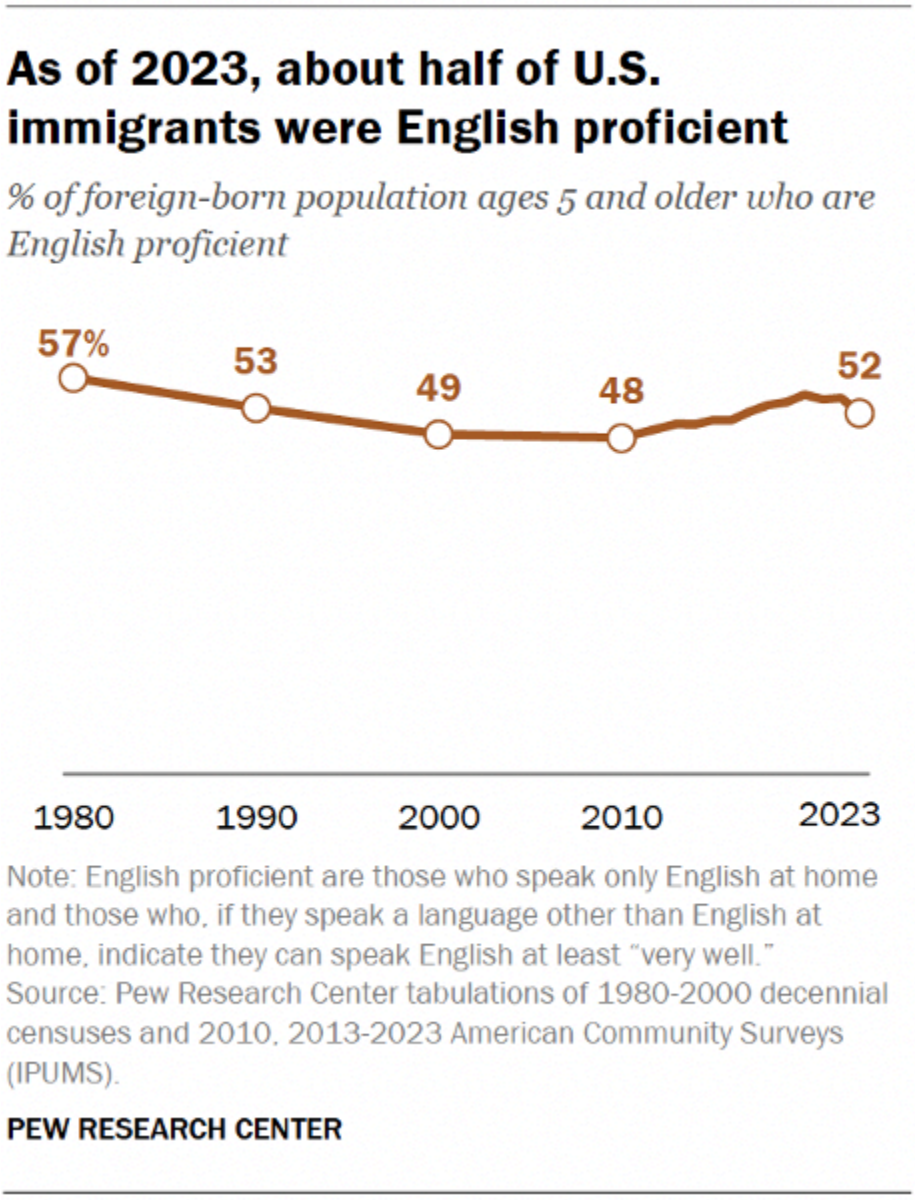


Figure 3. "As of 2023, about half of U.S. immigrants were English proficient." Pew Research Center, Washington, D.C. (2025, August 21).

🎬 Explore more: Video: Helping a Foreign-Born Patron Transfer Credentials (~2 min)

Libraries Bridging the Gap

Free access to books, films, and digital materials aids language acquisition and information literacy (Wang,

Huang, Li, & Chen, 2020). Many libraries partner with immigrant service organizations to connect patrons to legal, educational, and financial resources.

- **Marshall Public Library** hosts a multi-lingual story hour for children
- **Hennepin County Library** offers language practice with their English Conversation Circle held throughout the month
- **Toronto Public Library** offers free computer classes and settlement support for newcomers

These examples demonstrate how libraries act as bridges—providing technology, skills, and social connection that empower immigrants to thrive.

Reflection

Libraries are pillars of social equity. Bridging the digital divide requires continuous assessment and cultural awareness.

What barriers might immigrants face in your library's digital spaces?

How can you adapt programming to reflect your community's linguistic and cultural realities?

By centering on empathy, access, and adaptability, library workers ensure that digital equity extends to all who walk through their doors.

Additional Resources

Immigration Support: Know your rights

Resource	Description
American Civil Liberties Union	Know your rights as an immigrant
Immigrant Legal Resource Center	Downloadable & printable materials/resources
National Immigrant Justice Center	Information with downloads & examples

Patron Support: Remote courses

Resource	Description
LA Public Library – Cybernauts	Computer and mobile tech support
Seattle Public Library	Online resources
Free Library of Philadelphia	Online resources in language, health, and more

Program Ideas: Get inspired

Library Program	Description
Free Library of Philadelphia—Edible Alphabet	English Language Learning in a culinary setting
Toronto Public Library—Settling in Toronto	Assistance getting started in Toronto & Canada
Toronto Public Library—Computer & Library Training	Free, in-person classes offered throughout Toronto
Northwestern University—World Relief Chicagoland Summer Program	Makerspace program for immigrants

Key Takeaway

Immigrants may encounter barriers to digital participation related to language, access, and familiarity with online systems. Libraries support these users by providing technology access, relevant resources, and assistance navigating digital services.

Works Cited

American Library Association. (2006, June 30). *Library Bill of Rights.* American Library Association. https://www.ala.org/advocacy/intfreedom/librarybill

Cherewka, A. (2020). The Digital Divide Hits U.S. Immigrant Household Disproportionately during the COVID-19 Pandemic. *Migration Information Source*. https://www.migrationpolicy.org/article/digital-divide-hits-us-immigrant-households-during-covid-19

Geraghty, L., Panettieri, A., Pedrosa, S., & Luna, D. (2021). State of the Digital Divide in the

Hispanic Community. *National League of Cities*. https://www.nlc.org/wp-content/uploads/2021/11/HELO-POLICY-BRIEF-1.pdf

Hennepin County Library. (n.d). *Events*. Hennepin County Library, https://hclib.bibliocommons.com/v2/events?types=56f9a197414af7d259046a7f.

Igarashi, T., Koizumi, M., & Widdersheim, M. M. (2023). Overcoming social divisions with the public library. *Journal of Documentation, 79*(1), 52–65. https://doi.org/10.1108/JD-12-2021-0244

KGUN9. (2018, March 7). *County libraries help refugees, immigrants get degrees recognized in U.S* [Video]. YouTube. https://www.youtube.com/watch?v=TPc3aRGbh00

Kramer, S., & Passel, J. S. (2025, August 21). What the data says about immigrants in the U.S. *Pew Research Center*. https://www.pewresearch.org/short-reads/2025/08/21/key-findings-about-us-immigrants/

Ndumu, A. [CHOICE Media Channel]. (2025, August 1). *U.S. Newcomers in Academic Libraries: What Should We Know, What Can We Do* [Video]. YouTube. https://youtu.be/0WE4wOSzA1I?si=M7N2H9XxlpvYTSxF

New Americans Initiative. (n.d.). Los Angeles Public Library. Retrieved October 25, 2024, from https://www.lapl.org/newamericans

Sánchez, P., & Salazar, M. (2012). Transnational Computer Use in Urban Latino Immigrant Communities: Implications for Schooling. *Urban Education, 47*(1), 90–116. https://doi.org/10.1177/0042085911427740

U.S. Department of Labor. (2025). *Foreign-Born Workers: Labor Force Characteristics—2024*. https://www.bls.gov/news.release/pdf/forbrn.pdf

Wang, C., Huang, R., Li, J., & Chen, J. (2020). Towards better information services: A framework for immigrant information needs and library services. *Library & Information Science Research, 42*(1), 101000. https://doi.org/10.1016/j.lisr.2019.101000

West, J. C. (2011). *Without A Net: Librarians Bridging the Digital Divide*. Libraries Unlimited.

Wood, J. [Idaho Commission for Libraries]. (2024, February 13). *Building Community through a Multilingual Storytime* [Video]. YouTube. https://www.youtube.com/watch?v=skmF4N-TR-o

14.

DIGITAL DIVIDE AND INDIGENEITY

Nichole Colsch; Emily Lucie; and Grace Ridnour-Nollsch

Learning Objectives

By the end of this chapter, readers will be able to:

- understand how the digital divide affects Indigenous communities, including access to high-speed internet and technology hardware;
- define the middle mile and last mile; and
- identify ways state, local, and Tribal libraries can provide access.

The Long, Rural Road: Driving Toward Digital Equity in Indigenous Communities

In the 21st century, it is easy to assume everyone has equal access to something as simple as the internet. However, this is not the case. This chapter will explore the digital differences Indigenous communities in the United States face.

Due to Western Expansion and colonization, Indian Country in the United States is spread across 334 reservations in 35 states and 100 million acres of land (Delmar, 2023). Therefore, while this chapter may refer to Indigenous peoples as one whole group, it is important to recognize there are 574 federally recognized Native nations who may all have different digital needs. Dr. Traci Norris, the executive director of the American Indian Policy Institute (AIPI), brings attention to the difficulty Native Americans face when it comes to access to broadband.

YouTube Video: "Digital Inclusion Training Preview: Tribal Broadband Overview"





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Considering the vastness of Native nations that share geography with the United States, bridging the digital divide is a tremendous task. Research has shown repeatedly that Indigenous communities have less internet access, slower internet speed, and higher internet prices than surrounding areas which creates a barrier to achieving digital equity.

Many Miles to Go—How Wide is the Digital Divide in Indigenous America?

A key aspect of the digital divide is the lack of access to high-speed internet. Here are some key data points demonstrating some of these issues:

- 18% of reservation residents have no internet at all ("Exploring the Lack of Internet Access," 2023)
- 49% of reservation residents with no internet were using land-based internet (Morris & Howard, 2019)
- Download speeds are 75% slower on Tribal lands ("Exploring the Lack of Internet Access," 2023)

Looking at these statistics together, it's clear that access to high-speed internet is not increasing as much as it needs to be for Indigenous communities, leading to greater digital inequality.

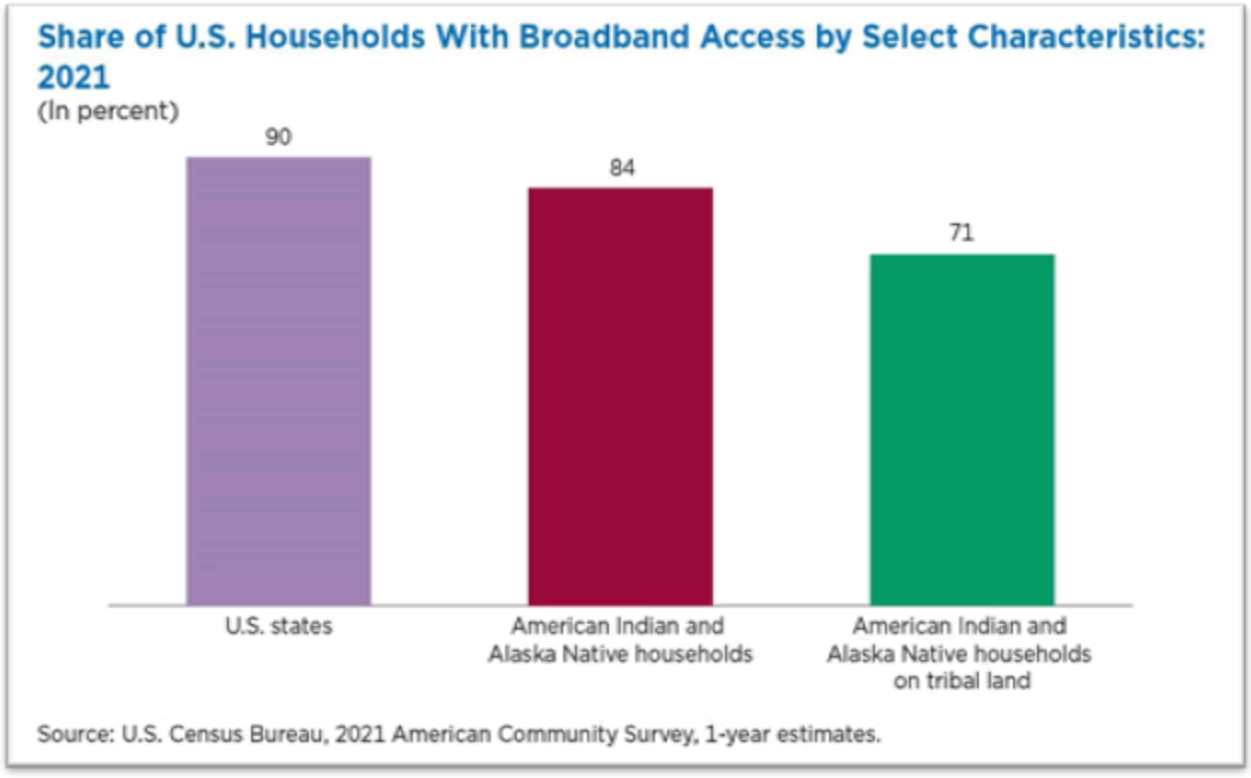


Figure 1. "Share of U.S. Households with Broadband Access by Select Characteristics: 2021 (In percent)." U.S. Census Bureau (2024, June 18).

But Wait! It's Not Just about Broadband?

High-speed internet is a large part of the digital divide, but lack of technology hardware is also a contributor. The AIPI study found that 33% of people living on reservations rely on smartphones to access the internet rather than computers. Many websites and programs are not mobile-friendly, so people relying on smartphones to access the internet cannot do so to its full capacity. Connectivity is also an issue with 31% of residents reporting spotty connection or no connection at all at home (Morris & Howard, 2019).

Rural Broadband + Poverty = The Expensive "Middle Mile"

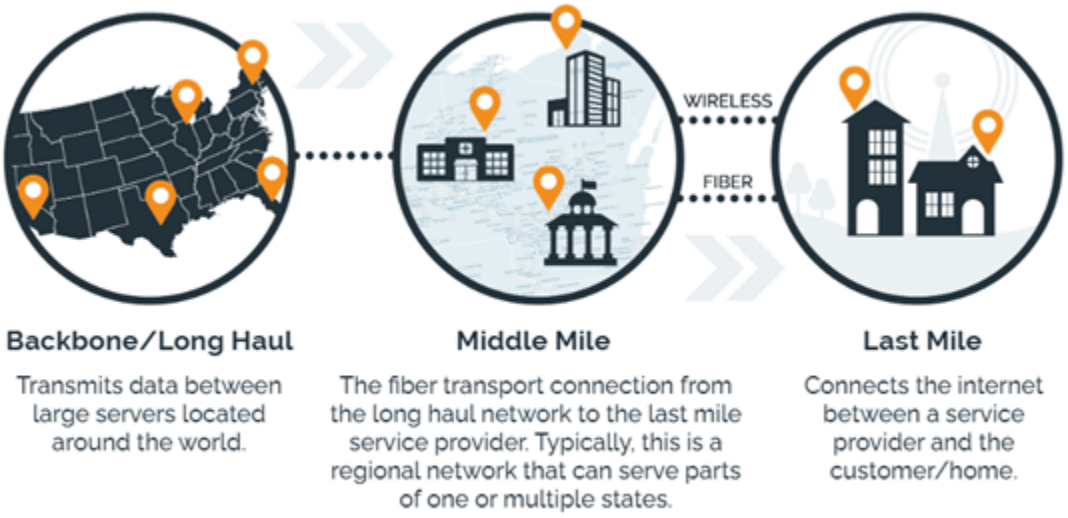


Figure 2. "Middle Mile Network: What is it and Why Does it Matter?" Win Technology (2023, January 18).

Some of the greatest obstacles to internet access are poverty and living in rural areas. These factors greatly affect Indigenous communities. Internet costs are 11% higher for Indigenous communities than for the public. This is largely because service providers are charging extra to provide internet to smaller, more rural communities like reservations ("Exploring the Lack of Internet Access," 2023). A 2016 survey found that 26.2% of Native people live in poverty. Therefore, around a quarter of the Native community may be relying on free services, which are often less available in rural communities. This is where public libraries and other organizations come in.

Bridging the Middle Mile: How Libraries & Advocacy Are Helping Close the Gap

Getting expensive rural broadband services to remote reservations and communities has complex challenges. Native nations have varying relationships with their state and local surroundings. Jurisdictional issues often complicate the administration of programs intended to serve them. Federal funds administered through state-run systems can be held up by administrative conflict (Delmar, 2023). Thus, addressing the digital divide's impact on Indigenous communities requires multi-pronged efforts.

State, local, and Tribal libraries combat the digital divide by providing free public broadband to local users. Public library systems enjoy the benefit of federal "e-rate" laws that reduce costs, but, until recently, Tribal libraries were not eligible for these rate reductions. By January 2022, advocates including the American Library

Association, American Indian Policy Institute, and Institute of Museum and Library Services had pressured the Federal Communications Commission to allow Tribal libraries to qualify for e-rate broadband pricing. This has allowed Tribal libraries to provide free broadband access to more Native users (Schwayder, 2020).

With the use of Internet2's Toward Gigabit Libraries Toolkit, Tribal libraries like Jemez Pueblo Community Library and Archives in New Mexico were able to improve their broadband internet for library users (Landgraf, 2022).

Libraries also work toward digital inclusion for Indigenous communities by administering grant-funded programs to expand broadband access (Ford, Slaughter, Neal, & Clark, 2024). COVID-19 era legislation drove significant expansion in federal funding driving what is referred to as a "Digital New Deal" for Native communities (National Digital Inclusion Alliance, 2023). A wide range of grants funded at the federal, state, and local levels are administered to meet local needs in Indigenous and other underserved populations.

The Last Mile: Toward Indigenous Digital Sovereignty

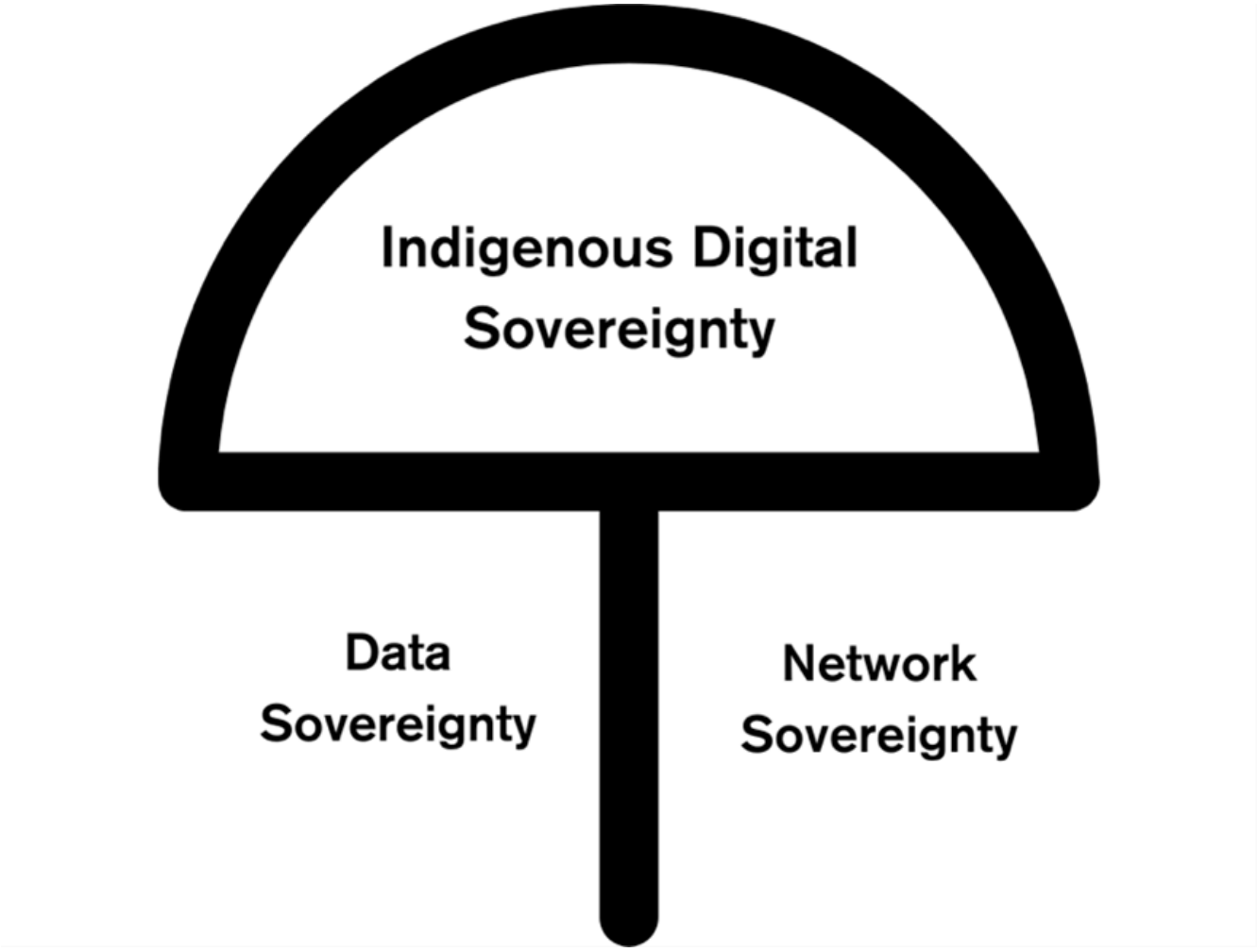


Figure 3. Indigenous Digital Sovereignty Defined

Academic and policy organizations also advocate for digital inclusion and **sovereign autonomy** for Indigenous communities in the United States. Arizona State University's American Indian Policy Institute recently announced the launch of the Center for Tribal Digital Sovereignty (Liese, 2024). The CTDS supports research on digital access, builds capacity, collects and disseminates information, and supports Indigenous digital sovereignty over both intangible data and physical hardware and networks ("Center for Tribal Digital Sovereignty," n.d.).

Go the Extra Mile

If you are interested in this topic, check out this presentation by Dr. Traci Morris, Executive Director of the American Indian Policy Institute for more information.



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Key Takeaway

Indigenous communities may face distinct technological challenges shaped by geography and infrastructure. Libraries benefit from understanding these contexts and supporting technology services that align with local needs and conditions.

Works Cited

Bauer, A., Feir, Donn. L., & Gregg, M. T. (2022). The tribal digital divide: Extent and Explanations. *Telecommunications Policy*, *46*(9), 102401. https://doi.org/10.1016/j.telpol.2022.102401

Center for Tribal Digital Sovereignty | American Indian Policy Institute. (n.d.). Arizona State University. Retrieved October 25, 2024, from https://aipi.asu.edu/ctds

Deerinwater, J. (2023). Crip Indigenous Storytelling across the Digital Divide. In M. Mills & R. Sanchez (Eds.), *Crip Authorship* (pp. 350–354). NYU Press; JSTOR. http://www.jstor.org/stable/jj.13944206.39

Delmar, D. (2023, July 19). *Indigenous Digital Sovereignty: From the Digital Divide to Digital Equity*. National Digital Inclusion Alliance. https://www.digitalinclusion.org/blog/2023/07/19/indigenous-digital-sovereignty/

Exploring the Lack of Internet Access on Native American Reservations. (2023, July 28). *Community Tech Network*. https://communitytechnetwork.org/blog/exploring-the-lack-of-internet-access-on-native-american-reservations/

Ford, B., Slaughter, A., Neal, J., & Clark, L. (2024, September 4). *Expand Library Digital Equity Services with New Funds Now! | Public Library Association*. American Library Association. https://www.ala.org/pla/education/onlinelearning/webinars/ondemand/expand

Hall, T. D. (2021). Information Redlining: The Urgency to Close the Digital Access and Literacy Divide and the Role of Libraries as Lead Interveners. *Journal of Library Administration*, *61*(4), 484–492. https://doi.org/10.1080/01930826.2021.1906559

Landgraf, G. (2022, March 1). *The Last Mile*. American Libraries Magazine. https://americanlibrariesmagazine.org/2022/03/01/the-last-mile/

Liese, S. (2024, June 10). First-of-its-kind digital sovereignty center for tribal nations launches [High Plains Public Radio]. *HPPR News*. https://www.hppr.org/hppr-news/2024-06-10/first-of-its-kind-digital-sovereignty-center-for-tribal-nations-launches

Mejía, D. (2024, June 18). American Indian and Alaska Natives in Tribal Areas Have Among Lowest Rates of High-Speed Internet Access. *United States Census Bureau*. https://www.census.gov/library/stories/2024/06/broadband-access-tribal-areas.html

Morris, T. (2023, July 14). Indigenous Digital Sovereignty Defined | American Indian Policy Institute. *ASU Sandra Day O'Connor College of Law – AIPI Blog*. https://aipi.asu.edu/blog/2023/07/indigenous-digital-sovereignty-defined

Morris, T., & Howard, B. (2019). Tribal Technology Assessment: The State of Internet Service on Tribal Lands. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3427547

National Digital Inclusion Alliance (NDIA) (Director). (2023, July 12). *Indigenous Digital Sovereignty: From the Digital Divide to Digital Equity* [Video recording]. https://www.youtube.com/watch?v=D1LIBoOZ79c

Shwayder, M. (2020, September 17). Thanks to Libraries, Remote Tribes Are Now Getting Online | Digital Trends. *Digital Trends*. https://www.digitaltrends.com/news/libraries-municipal-fiber-broadband-new-mexico-native-reservation/

Tribal Libraries Now Have Access to Federal Funds for Affordable Internet Under FCC E-Rate

Program. (2022, January 27). Institute of Museum and Library Services. http://www.imls.gov/news/tribal-libraries-now-have-access-federal-funds-affordable-internet-under-fcc-e-rate-program

Toward Gigabit Libraries. (n.d.). Internet2. Retrieved November 4, 2024, from https://internet2.edu/community/community-anchor-program/cap-library-resources/toward-gigabit-libraries/

15.

DIGITAL DIVIDE AND RACE

Adrienne Combs; Isa Maginnis; and Victor Resendiz

Learning Objectives

By the end of this chapter, readers will be able to:

- understand how the digital divide affects people of color;
- learn how factors like the education system and the COVID-19 pandemic contribute to the digital divide; and
- identify strategies and tools libraries can use to address the digital divide and provide access to users.

The Digital Divide Has Disproportionately Affected People of Color.

The digital divide is a systemic bias that has historically impacted people of color more severely than compared to White people. As Tracy Hall points out, people of color are often economically disenfranchised, making it difficult for them to afford home broadband internet access and personal technology (Pun, 2021). This economic inequality exacerbates the digital gap. A study by Kienbaum in 2020 on racial disparities in digital access found that "children in one of every three Black, Latinx, and Native American households [in the United States] did not have broadband access at home" (Pun, 2021).

A study from Pew Research Center shares that compared to 80% of White adults, 69% of Black adults and 67% of Hispanic adults have access to a desktop or laptop computer. Compared to the same 80% of White adults, 71% of Black adults and 65% of Hispanic adults have broadband internet (Atske & Perrin, 2021).

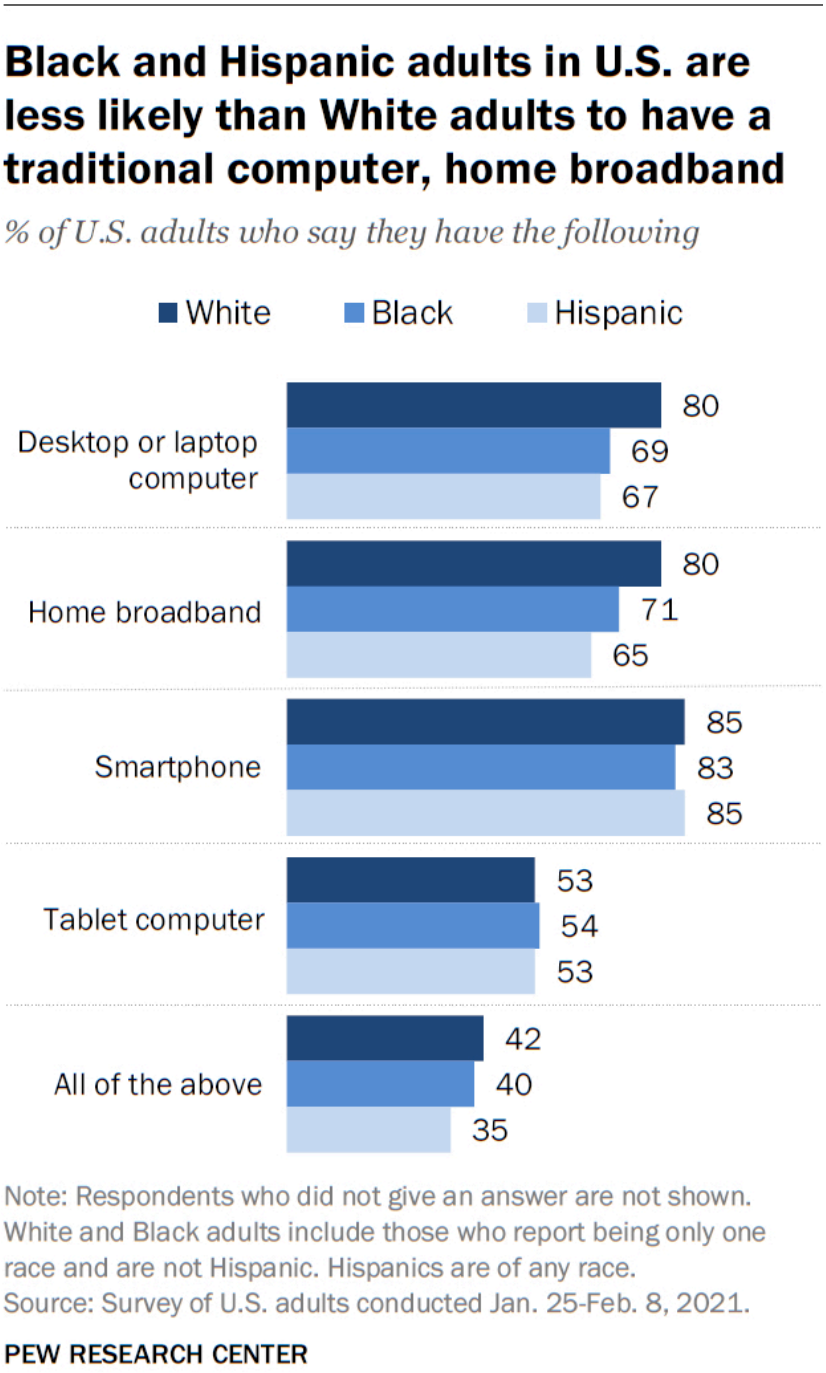


Figure 1. "Black and Hispanic adults in U.S. are less likely than White adults to have a traditional computer, home broadband." Pew Research Center, Washington, D.C. (2021, July 16).

This lack of access creates significant obstacles for BIPOC individuals in various socio-economic areas, including job seeking and retention. As more jobs require consistent internet access, and considering how many have come to rely on technology in the wake of the COVID-19 pandemic, those without it are left at

a disadvantage. Furthermore, limited access to technology complicates educational efforts, further hindering socio-economic progress. The inability to consistently access digital tools also disrupts the right to intellectual freedom, a fundamental right to which all citizens are entitled. While many BIPOC individuals rely on library services to bridge the digital gap, these resources often come with their own set of challenges. Barriers such as transportation issues, limited availability of public computers or devices, and short time limits for computer use create further difficulties. Though essential, reliance on public resources is not always sufficient to close the digital divide.

As technology and the internet become increasingly central to success in education, work, and personal development, addressing this digital disparity is critical to promoting equal opportunities for all. This digital divide is further perpetuated within the public school system, where resources and technology are often unevenly distributed across schools and communities.

The Digital Divide in Schools: How the COVID-19 Pandemic brought us back to square one—and how educators are swamped with technology now.

The education system plays a role in the digital divide not only through the technology standards States choose to teach to students but also by validating (or not validating) student use of computers and digital spaces. Educators should be asking how students (especially students of color) want to use computers and digital spaces for their education, but often the problem of the digital divide in schools is a biased interpretation of data that ignores how and when Black and Latino students choose to make digital spaces their own––and that makes overall technology use seem like an 'ineffective' tool in the classroom (Tichavakunda & Tierney, 2018). Classroom technologies that are perceived as 'ineffective' make quite the convenient argument for not investing in technologies for public education, among others, such as how introducing tech to curriculum might advantage the already advantaged (O'Brien & Scharber, 2008). And this would have been the tone of the digital divide problem in schools—had it not been for COVID-19 and the post-pandemic classroom.

A call to action for technology and refined social media (and video games) has become the standard message of upscaling technology—especially laptops—in public education nationwide. Most of this momentum can be attributed to the "homework gap," our present manifestation of the digital divide in which students of color without adequate computers or internet failed to meet homework requirements during and after the pandemic (Anderson, 2022).

The response to the call has been a mix of Roblox Education, more hybrid learning, and the continued question of what to do about student cell phones (Hatfield, 2024; Herold, 2023). Teen cell phone use has also entered the digital divide with its own new problem: how do we understand the student who does not own a laptop but owns a smartphone and knows the who's who of gaming, streaming, and e-book reading, but they might not have a real way to turn in digital homework?

And there is also the question of "The Second Digital Divide," how students of color and their educators can capitalize on the use of student-made digital spaces, like social media platforms, to benefit academics (Tichavakunda & Tierney, 2021). And with that, will the problem of students' 'effective' use of technology return once access to computers reaches a new post-pandemic equilibrium?

Most likely.

How Libraries Can Bridge the Gap.

Libraries are a vital tool in bridging the digital divide, access to the internet and computers are free to patrons within a library––and sometimes it's the only place they have such access. Libraries are doing what they can to provide this with the resources they have, but this access can be heavily dependent on how many computers libraries have at their disposal, the hours they're open, and time restrictions amongst other things (MacKenzie, 2021).

There is a high demand for technology from libraries, especially public libraries. Because they are a public institution they are susceptible to budget cuts in less wealthy areas. Which greatly impacts the already limited access some of these communities have to technology and the internet. Libraries are doing their best to step up to the challenge of the digital divide, with 47% of libraries offering Wi-Fi hotspots for patrons to check out––providing their communities with vital internet access (Goek, Jones, & Thompson, 2024). Combined with the standard that libraries will assist with the technology provided, resulting in 95% of public libraries offering training within the realm of digital literacy (Goek, Jones, & Thompson, 2024).

But there is always more that can be done, especially in regards to keeping libraries open in underserved communities, which is dependent on sustained effort from the government that provides funding, and librarians and patrons whose use of a library encourages the continued support from said government (Bill and Melinda Gates Foundation & AARP, 2004). Programs that offer Wi-Fi hotspots could be implemented––when libraries have the funds to do so––a broader range of languages could be implemented within the technology libraries provide for easier access for patrons. There are many tools libraries can implement to help bridge the digital divide.

Key Takeaway

Differences in technology access and digital skills are influenced by broader social and economic patterns. Libraries help mitigate these effects by providing consistent access to technology, instruction, and support.

Works Cited

Anderson, M., Faverio, M., & McClain, C. (2022, June 2). How Teens Navigate School During COVID-19. *Pew Research Center*. https://www.pewresearch.org/internet/2022/06/02/how-teens-navigate-school-during-covid-19/

Atske, S., & Perrin, A. (2021, July 16). Home broadband adoption, computer ownership vary by race, ethnicity in the U.S. *Pew Research Center*. https://www.pewresearch.org/short-reads/2021/07/16/home-broadband-adoption-computer-ownership-vary-by-race-ethnicity-in-the-u-s/

Bill & Melinda Gates Foundation & AARP. (2004). *Toward Equality of Access the Role of Public Libraries in Addressing the Digital Divide*. Bill & Melinda Gates Foundation. Retrieved from the Library of Congress, https://lccn.loc.gov/2005551815

Goek, S., Jones, K., & Thompson, J. (2024, July 30). *Technology in Public Libraries: Results from PLA's 2023 Annual Survey | Public Library Association*. American Library Association. https://www.ala.org/pla/education/onlinelearning/webinars/ondemand/2023techresults

Hatfield, J. (2024, June 12). 72% of U.S. high school teachers say cellphone distraction is a major problem in the classroom. *Pew Research Center*. https://www.pewresearch.org/short-reads/2024/06/12/72-percent-of-us-high-school-teachers-say-cellphone-distraction-is-a-major-problem-in-the-classroom/

Herold, B. (2023, May 15). Roblox Makes Its Move Into STEM Education. What Are the Risks for Schools? *Education Week*. https://www.edweek.org/teaching-learning/roblox-makes-its-move-into-stem-education-what-are-the-risks-for-schools/2023/05

MacKenzie, K. (2021). Public Libraries Help Patrons of Color to Bridge the Digital Divide, but Barriers Remain. *Evidence Based Library and Information Practice*, *16*(4), 138–140. https://doi.org/10.18438/eblip30035

O'Brien, D., & Scharber, C. (2008). Digital Literacies Go to School: Potholes and Possibilities. *Journal of Adolescent & Adult Literacy*, *52*(1), 66–68. JSTOR.

Pun, R. (2021). Understanding the Roles of Public Libraries and Digital Exclusion Through Critical Race Theory: An Exploratory Study of People of Color in California Affected by the Digital Divide and the Pandemic. *Urban Library Journal*, *26*(2). https://academicworks.cuny.edu/ulj/vol26/iss2/1

Tichavakunda, A. A., & Tierney, W. G. (2018). The "Wrong" Side of the Divide: Highlighting Race for Equity's Sake. *The Journal of Negro Education*, *87*(2), 110–124. JSTOR.

16.

DIGITAL DIVIDE AND RURALITY

Darryl Moton; Brenna Mulford; Mariel Murray; and Chelsea Rider

Learning Objectives

By the end of this chapter, readers will be able to do the following:

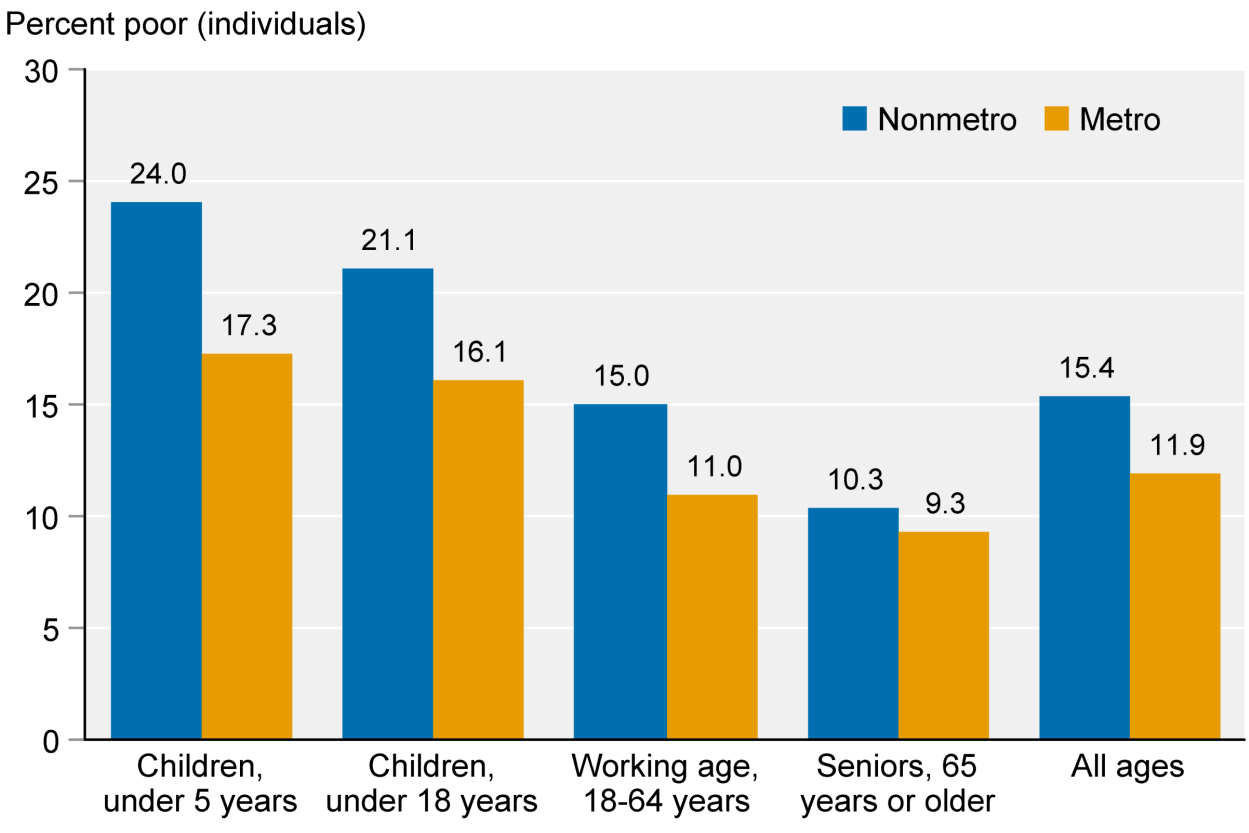
- articulate a clear, functional understanding of the definitions of "digital divide" and "rurality;"
- recognize how geographical and economic factors contribute to the digital divide, especially in analyzing how rural communities are impacted by price differentials and limited access to infrastructure; and
- evaluate the role of public libraries and initiatives in addressing this divide.

The digital divide is defined as "the gap between those who have affordable access, skills, and support to effectively engage online and those who do not" (NDIA, 2025). While the term "rurality" can have varied definitions based on the purpose of its usage, the U.S. Census Bureau defines "rural" as "any population, housing, or territory **not** in an urban area" (U.S. Department of Agriculture, n.d.).

The Problem

Much of the figurative "digital divide" can be traced to a more literal one––that of simple geography. A 2018 Pew Research Center survey found 24% of adults in rural areas were more likely to have difficulties gaining access to high-speed internet, compared to only 13% of adults in urban areas (Vogels, 2021). This is partly due to socioeconomic factors, such as poverty, age, and location, which create a lack of resources needed for proper access.

Poverty rates by age group and metro/nonmetro residence, 2019



Source: USDA, Economic Research Service using data from the U.S. Department of Commerce, Bureau of the Census, annual American Community Survey, 2019.

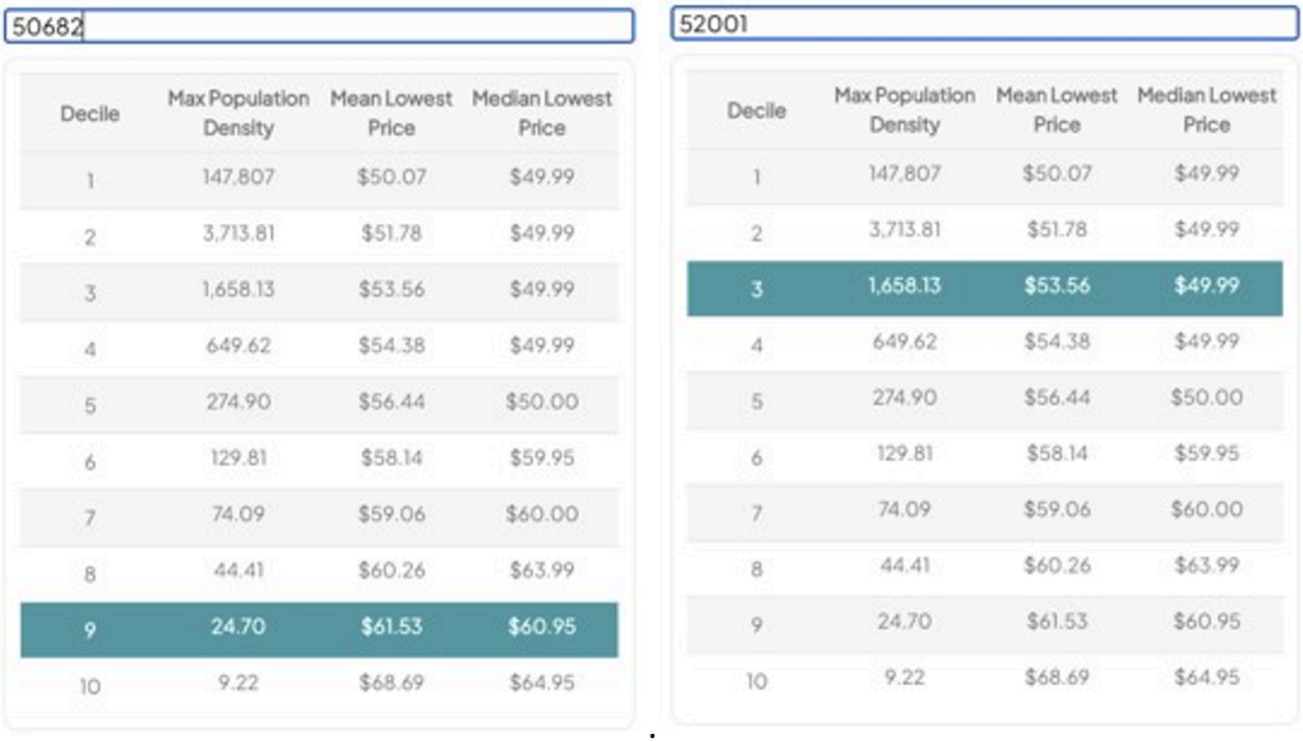
Figure 1. "Poverty rates by age group and metro/nonmetro residence, 2019." U.S. Department of Agriculture (2021, June 17).

The Cost of the Location Divide

Due to the market-based nature of broadband internet access, more far-flung locations disincentivize companies from investing in rural areas. This makes high-speed internet more expensive relative to distance from a major metropolitan area. Physical location already plays a significant role in population construction, wages, and access to public services such as libraries, schools, and other information centers; it also directly impacts how information reaches homes themselves. Digital Planet, an ongoing Tufts University research initiative, aggregates data relating physical location to broadband affordability and access. Utilizing data from, among other sources, the U.S. Census Bureau and the internet consumer reports service Broadband Now, Digital Planet provides visual representation of those costs.

Below are two communities, one rural and one urban. The two charts from Broadband Now show the

difference in broadband prices and population density per square mile by zip code (Torng, 2022). These are the lowest prices offered by any provider to install wired broadband. The charts show that prices in the 52001 zip code—Dubuque, Iowa, an urban area—are much cheaper compared to those in the zip code of 50682—Winthrop, Iowa, a rural community. Location plays a part in population, wages, and accessibility for libraries, information centers, and family.



50682

Decile	Max Population Density	Mean Lowest Price	Median Lowest Price
1	147,807	$50.07	$49.99
2	3,713.81	$51.78	$49.99
3	1,658.13	$53.56	$49.99
4	649.62	$54.38	$49.99
5	274.90	$56.44	$50.00
6	129.81	$58.14	$59.95
7	74.09	$59.06	$60.00
8	44.41	$60.26	$63.99
9	24.70	$61.53	$60.95
10	9.22	$68.69	$64.95

52001

Decile	Max Population Density	Mean Lowest Price	Median Lowest Price
1	147,807	$50.07	$49.99
2	3,713.81	$51.78	$49.99
3	1,658.13	$53.56	$49.99
4	649.62	$54.38	$49.99
5	274.90	$56.44	$50.00
6	129.81	$58.14	$59.95
7	74.09	$59.06	$60.00
8	44.41	$60.26	$63.99
9	24.70	$61.53	$60.95
10	9.22	$68.69	$64.95

Figure 2. Screen captures from Broadband Now, showing two charts that depict the difference in broadband prices and population density per square mile by zip code.

Impact of Public Libraries on the Rural Digital Divide

For those who lack internet access at home, libraries can be the only source of broadband internet access. On June 7th, 2019, the Public Library Association announced the DigitalLead: Rural Libraries Creating New Possibilities program ("Public Library Association Expands Access," 2019). This program used $400,000 of funding from Microsoft to provide digital resources and training to rural libraries, including hotspots ("DigitalLead," n.d.). An increasing number of libraries are adding hotspots to their circulating collections, and these continue to be in high demand, frequently having long holds lists (American Library Association, 2024). These can be a particularly effective tool in rural areas to counter the lack of infrastructure or resources which allow residents in-home internet access.



Figure 3. "Free WiFi Zone," by Charles Hutchins on Flickr is licensed under CC by 2.0.

In early 2020, while libraries in the United States closed their doors to the public due to the COVID-19 pandemic, Microsoft donated $120,000 to the Public Library Association to fund the addition of Wi-Fi access points outside of library buildings in rural areas. This allowed patrons to access Wi-Fi from a public resource while continuing to practice social distancing ("PLA and Microsoft Public WIFI Access Micro Grant Program," 2020). A 2019 grant through the Institute of Museum and Library Services allowed communities to use broadband Wi-Fi through "unused parts of the television broadcast spectrum." The first phase made it so five libraries could install "TV white space base stations" in their communities which let them extend library Wi-Fi beyond the actual library building (Landgraf, 2019).

Beyond the physical access that libraries provide for digital resources (e.g. Wi-Fi, hotspots, and public access computers), public libraries can also provide user support in their rural communities. Although urban libraries tend to offer a greater volume and variety of support options commensurate with their larger staff and budgets, small rural libraries also promote technology training and resources that focus on technology skill development (Gregg & McKendry, 2019). Promoting the use of technological resources and providing education through programming and one-on-one help sessions, libraries of all sizes play a role in ensuring equitable access to information.

Looking Ahead

In 2022, the entire United States—including all 50 states, Washington, D.C., Puerto Rico, American Samoa, Guam, the U.S. Virgin Islands, and the Commonwealth of the Northern Mariana Islands—were invited to submit a letter of intent to participate in the Broadband Equity, Access, and Deployment (BEAD) program. Following the letter of intent, each "Eligible Entity" then had the option to submit an application for Initial Planning Funds. On June 30th, 2023, the National Telecommunications and Information Administration announced the amount of funding each Eligible Entity would receive of the $42.45 billion grant ("Broadband Equity, Access, and Deployment Program," n.d.).



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BEAD prioritizes areas considered "unserved" (without access to 25/3 Mbps) and "underserved" (without access to 100/20 Mbps). Implementation began in 2024 and will take place over an estimated four-year time period ("Broadband Equity, Access, and Deployment Program," n.d.).

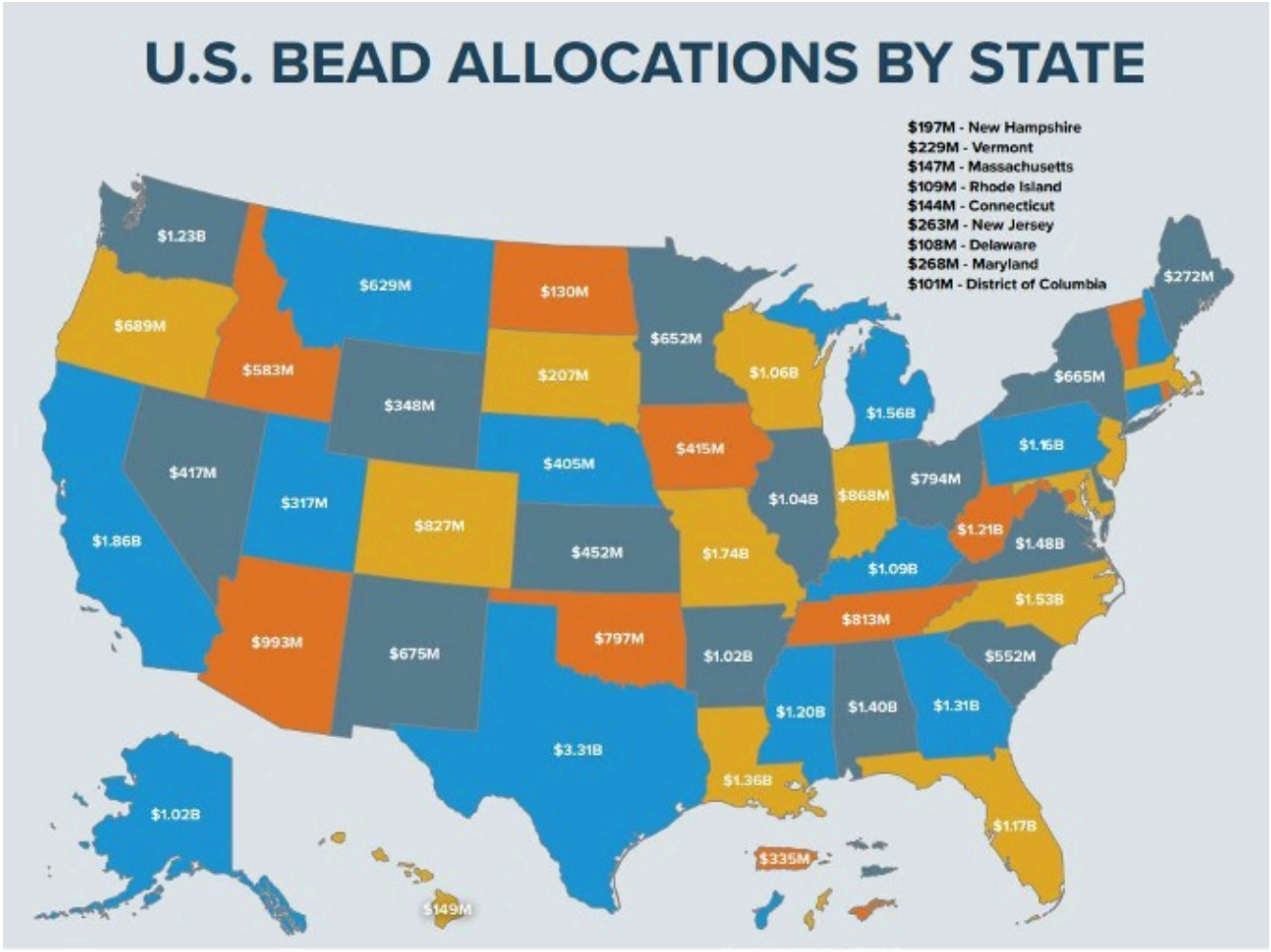


Figure 4. "U.S. BEAD allocations by State." Fiber Net Door Company (n.d.).

Some of the ways Eligible Entities may use their funding include fiber optic cable routing ("Bead Allocation Methodology," n.d.), training programs, and Wi-Fi installation in residential facilities ("Broadband Equity, Access, and Deployment Program," n.d.).

Key Takeaway

Rural communities often experience limitations in broadband access and technical infrastructure. Libraries serve as critical access points by offering connectivity, devices, and assistance for everyday digital tasks.

Works Cited

6 charts that illustrate the divide between rural and urban America. (2017, March 17). PBS News. https://www.pbs.org/newshour/nation/six-charts-illustrate-divide-rural-urban-america

American Library Association. (2024, July 9). *New Public Library Technology Survey report details digital equity roles*. ALA. https://www.ala.org/news/2024/07/new-public-library-technology-survey-report-details-digital-equity-roles

BEAD Allocation Methodology. (n.d.). Internet for All. Retrieved October 26, 2024, from https://www.internetforall.gov/program/broadband-equity-access-and-deployment-bead-program/bead-allocation-methodology

Broadband Equity, Access, and Deployment (BEAD) Program. (n.d.). Internet for All. Retrieved October 26, 2024, from https://www.internetforall.gov/program/broadband-equity-access-and-deployment-bead-program

Definitions – National Digital Inclusion Alliance. (2025, May 1). National Digital Inclusion Alliance. Accessed on Sept 29, 2025 from https://www.digitalinclusion.org/definitions/

DigitalLead: Rural Libraries Creating New Possibilities | Public Library Association. (n.d.). American Library Association. Retrieved October 26, 2024, from https://www.ala.org/pla/initiatives/digitallead

Digital Planet (2025). Broadband Infrastructure Funding and the Digital Divide: Prioritizing People Over Miles. *Tufts University*. Retrieved October 6, 2025. https://digitalplanet.tufts.edu/broadband-infrastructure-funding-and-the-digital-divide-prioritizing-people-over-miles/

Grant Information | Door County, WI. (n.d.). FiberNet Door County: Broadband Infrastructure Project. Retrieved October 26, 2024, from https://www.co.door.wi.gov/1175/Grant-Information

Gregg, A., & McKendry, V. (2019). Encouraging essential skills: How public libraries promote technological literacy in rural communities. *Journal of New Librarianship, 4*(2), 476–504. https://doi.org/10.21173/newlibs/8/2

Hutchins, C. (2011). *Free wifi zone* [Photo]. https://www.flickr.com/photos/celesteh/6237633144/

Landgraf, G. (2019, May 1). Wi-Fi in the "White Space." *American Libraries Magazine*. https://americanlibrariesmagazine.org/2019/05/01/wi-fi-digital-white-space-libraries/

Lee King, D. (2021). Chapter 4: The Digital Divide. *Library Technology Reports, 57*(2), 24–28.

PLA and Microsoft Public WiFi Access Micro Grant Program. (2020, April 20). American Library Association. https://www.ala.org/pla/initiatives/publicwifi

Public Library Association expands access to technology and digital skills training for rural communities with Microsoft Philanthropies. (2019, June 7). American Library Association. https://www.ala.org/news/2019/06/public-library-association-expands-access-technology-and-digital-skills-training-rural

Rural Poverty & Well-Being – Economic Research Sevice. U.S. Department of Agriculture (Jan 14 2025). Accessed on Sept 29 2025 from https://www.ers.usda.gov/topics/rural-economy-population/rural-poverty-well-being

Steele, C. (2018, November 16). Top Five Digital Divide Solutions. *Digital Divide Council*. http://www.digitaldividecouncil.com/top-five-digital-divide-solutions/

Torng, K. (2022, May 6). Digital Divide: Broadband Pricing by State, Zip Code, and Income 2019. *BroadbandNow*. https://broadbandnow.com/research/digital-divide-broadband-pricing-state-zip-income-2019

U.S. Census Bureau. (n.d.). *Urban and Rural*. Census.Gov. Retrieved October 26, 2024, from https://www.census.gov/programs-surveys/geography/guidance/geo-areas/urban-rural.html

U.S. Department of Agriculture. (n.d.) *What is Rural?* National Agricultural Library. https://www.nal.usda.gov/rural-development-communities/what-is-rural

Vogels, E. A. (2021, August 19). Some digital divides persist between rural, urban and suburban America. *Pew Research Center*. https://www.pewresearch.org/short-reads/2021/08/19/some-digital-divides-persist-between-rural-urban-and-suburban-america/

17.

DIGITAL DIVIDE AND SOCIOECONOMICS

Marion Gaughan; Lauren Hanssen; Allison Honz; Sidney Matzdorff; and Emily Ostrander

Learning Objective

Understand how socioeconomic factors—such as income, education, and employment—affect the digital divide. By assessing the factors present in their communities, librarians are better able to tailor programs and services to their patrons.

What is Socioeconomics and the Digital Divide?

The digital divide is the uneven access, affordability, and technology literacy for an individual or people of a specific group (Computers for Community, 2022). This imbalance can be due to social and/or geographical aspects. In this chapter, we will discuss the effects of socioeconomics on the digital divide. Socioeconomics compares one's position in a community based on income, education, and employment (See Figure 1).

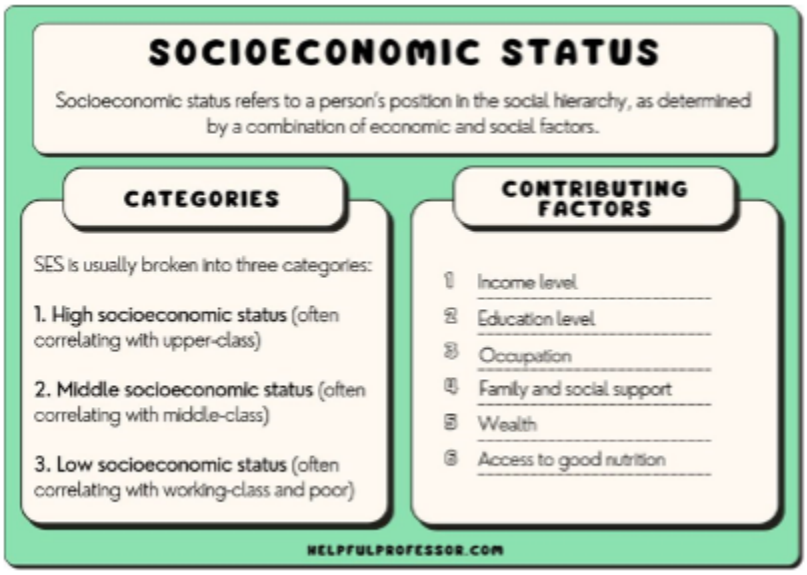


Figure 1. "Socioeconomic Status." HelpfulProfessor.com (2023, October 21).

The Problem with Accessibility and Affordability

Socioeconomic status has the ability to influence technology access. Rural households or communities in impoverished areas face barriers when it comes to accessibility of technology (See Chapter 16). Local libraries in smaller, rural communities may not have the funding to provide adequate technology for patrons' needs. Lack of access to technology can greatly limit a household or community's ability to obtain employment opportunities, health information, and other essential resources.

These limitations also apply to urban and suburban areas, where technology is typically more readily available, but could still be inaccessible to some. People living in urban or suburban areas often have the resources available to access technology, however, due to lower incomes or lack of employment these groups are not able to afford internet or other services. This can negatively affect their opportunities for higher-paying jobs or remote job possibilities. These factors can all contribute to one's socioeconomic status (SES) and create a digital divide.

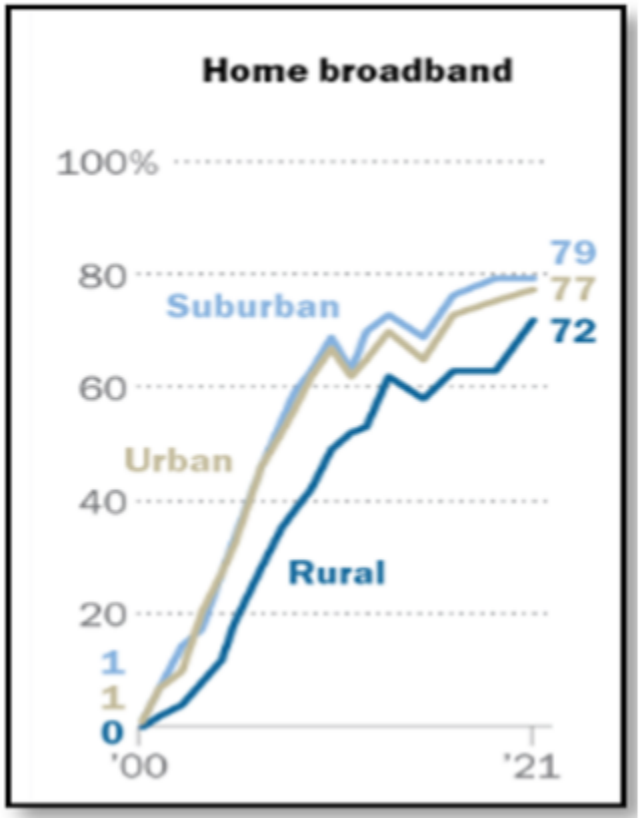


Figure 2. "Despite growth, rural Americans have consistently lower levels of technology ownership than urbanites and lower broadband adoption than suburbanites." Pew Research Center, Washington, D.C. (2021, August 17).

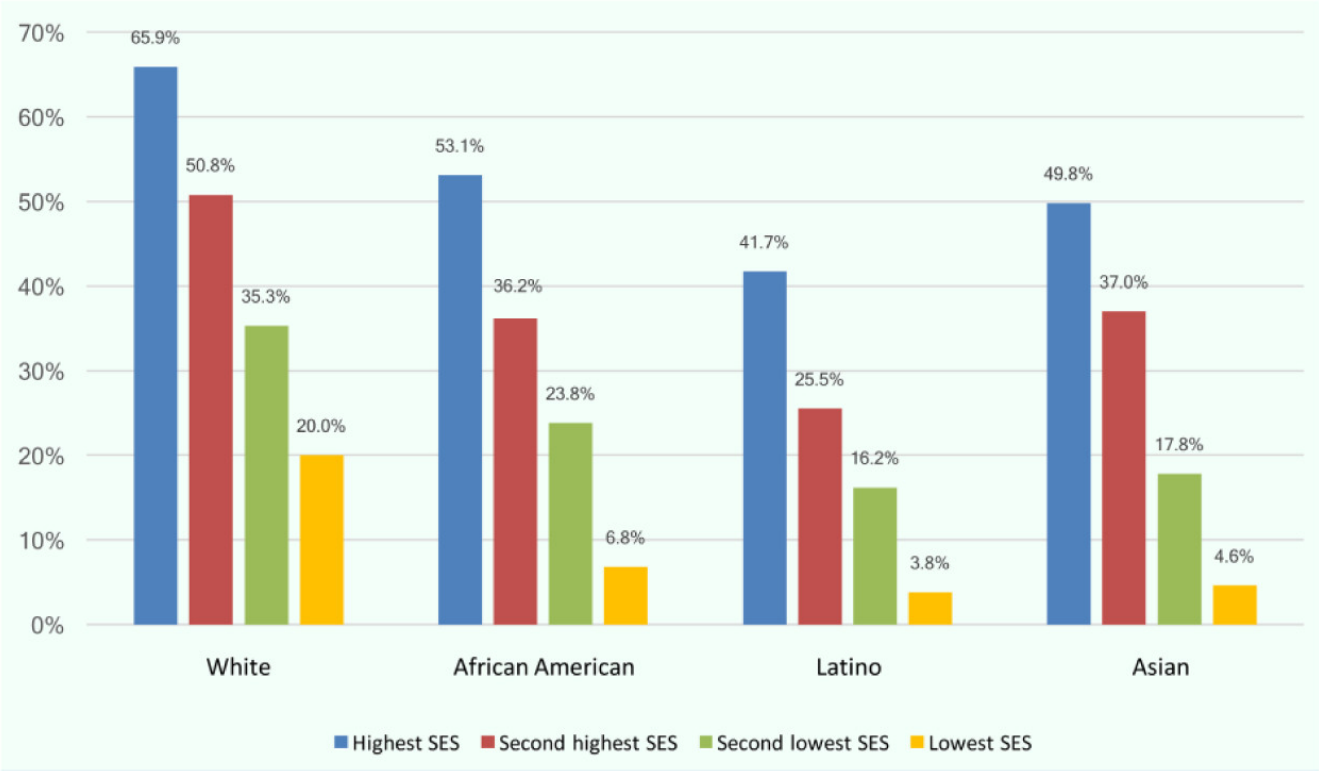


Figure 3. "Predictive probability of the internet use for health information by race/ethnicity and SES." Yoon, H., Jang, Y., Vaughan, P. W., & Garcia, M. (2020).

In July 2024, the FCC (Federal Communications Commission) approved using e-rate funding for hotspots and service contracts for libraries and schools. This ruling will allow more libraries to purchase hotspots for their patrons for home use, making broadband internet more accessible to individuals with lower incomes, especially in rural areas where internet service providers don't always reach. Some companies, such as T-Mobile and UScellular, offer hotspot devices and service contracts for as little as $25 per month per device to community libraries, easing the financial barrier for libraries seeking to bridge the accessibility and affordability divide on a local level.

The Problem with Digital Literacy

Digital literacy is the skill set needed to access and effectively use technology to complete tasks. As communities and schools strive to create more accessibility to technology, this highlights the problem of having the technology but not having the skill set to use it. Lev-On, Steinfeld, Abu-Kishk, & Pearl Naim (2021) found that there "is a significant social value in reducing digital inequality by increasing digital literacy of disadvantaged populations, i.e. citizens with low socioeconomic status and low digital literacy" (p. 146). This demonstrates the value of closing the digital divide and how people with a lower SES could be positively impacted.

Library systems can help bridge this divide by meeting patrons where they are and assessing the needs of their community in order to offer the technology and tech support most needed by their patrons. The Fulton County Library System in Atlanta, Georgia makes these services a priority, offering computer and Wi-Fi access, circulating Chromebooks to take home for a certain amount of time, and mobilizing their library resources through their "innovation stations" that bring library resources to communities who may have difficulty in traveling to the library. To supplement this greater access, the Fulton County Library System offers classes ranging in technology complexity from "Email for Beginners" to "Step into the Future: Virtual Reality Program," addressing the vastness of the digital literacy spectrum in their region.

Statistics on Low Socioeconomic Status and Technology Ownership

Studies on the correlation between low socioeconomic status and digital inequality have found many substantial links. A Carnegie Corporation of New York study found that 65% of students and their families who identify as part of a lower SES are negatively impacted by their inability to access a computer (Carnegie Corporation of New York, 2021). This lack of computer access can prevent students from the educational opportunities needed to succeed. Pew Research Center also identified the correlation between income level and technology ownership, finding that in every category, individuals of the lowest income bracket also owned the least amount of technology (See Figure 4).

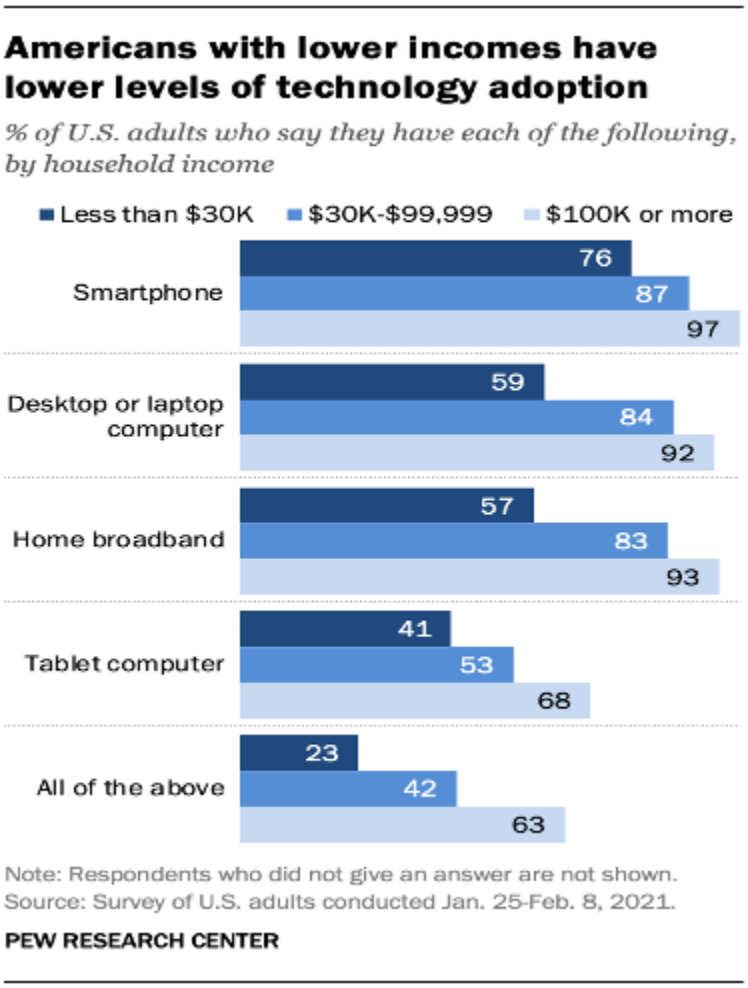


Figure 4. "Americans with lower incomes have lower levels of technology adoption." Pew Research Center, Washington, D.C. (2021, June 21).

Three years after this study, Pew Research Center's 2024 survey also found that, while Americans in the lowest income bracket (>$30,000/year) are least likely to own a smartphone, they are also the most likely of any socioeconomic group to be smartphone dependent (Gelles-Watnick, 2024). This results in individuals with lower SES being more likely to use smartphones for tasks traditionally reserved for larger screens, like job applications, schoolwork, health forms, etc. (Vogels, 2020).

Other Resources for Library Staff

The State Library of Iowa has digital literacy training for library staff members, including their continuing education video "We're All Tech Librarians: Digital Literacy" on YouTube. These programs and other digital learning opportunities, for both library users and staff members, are continuing to close the gap in digital literacy and accessibility. The ALA also frequently publishes news related to digital equity, and occasionally

compiles resources within its "Digital Literacy Task Force" for libraries seeking to close the divide. All of these resources are recommended for library staff to stay in the know, educated, and equipped to confront the socioeconomic component to the digital divide.

Key Takeaways

Socioeconomic conditions affect individuals' ability to access and use digital technologies. By providing public access to devices, connectivity, and instruction, libraries help reduce practical barriers to digital participation.

Works Cited

Lev-On, A., Steinfeld, N., Abu-Kishk, H., & Pearl Naim, S. (2021). The long-term effects of digital literacy programs for disadvantaged populations: Analyzing participants' perceptions. *Journal of Information, Communication and Ethics in Society, 19*(1), 146–162. https://doi.org/10.1108/JICES-02-2020-0019

Baig, E. C. (n.d.). FCC Expands Libraries' Wi-Fi Hot Spot Lending Program. *AARP*. Retrieved October 31, 2024, from https://www.aarp.org/home-family/personal-technology/info-2024/fcc-library-hot-spot-lending-program.html

Carnegie Corporation of New York. (2021, July 8). New Data Shows How Digital Inequality Affects Lower-Income Students and Their Families | Family & Community Engagement. *Carnegie Corporation of New York*. https://www.carnegie.org/our-work/article/new-data-shows-how-digital-inequality-affects-lower-income-students-and-their-families/

Community Connect: Fostering Digital Access | ALA. (n.d.). *American Library Association*. Retrieved October 31, 2024, from https://www.ala.org/tools/programming/communityconnect/fostering

Computers for Community. (2022, June 10). *The Digital Divide Explained* [Video]. YouTube. https://www.youtube.com/watch?v=m7PzD2m3ey4

Dawson, W. & Sidoti, O. (2024, November 13). Internet, Broadband Fact Sheet. *Pew Research Center*. https://www.pewresearch.org/internet/fact-sheet/internet-broadband/

Digital Inclusion Programs for Libraries | T-Mobile for Education. (n.d.). *T-Mobile for Education*. Retrieved October 31, 2024, from https://www.t-mobile.com/business/industry-solutions/libraries

Doran-Myers, M. (2017, August 2). ALA report finds that the number of rural public libraries offering internet services increases to 85%. *Library Research Service*. https://www.lrs.org/2017/08/02/ala-report-finds-that-the-number-of-rural-public-libraries-offering-internet-services-increases-to-85/

Drew, C. (November 4, 2022). 15 Socioeconomic Status Examples (Top Influencing Factors). *Helpful Professor*. https://helpfulprofessor.com/socioeconomic-status-examples/

Gelles-Watnick, R. (31 Jan. 2024). Americans' Use of Mobile Technology and Home Broadband. *Pew Research Center*. www.pewresearch.org/internet/2024/01/31/americans-use-of-mobile-technology-and-home-broadband/.

Hall, T. D. (2021). Information Redlining: The Urgency to Close the Digital Access and Literacy Divide and the Role of Libraries as Lead Interveners. *Journal of Library Administration, 61*(4), 484–492. https://doi.org/10.1080/01930826.2021.1906559

Lcom Team. (2023, January 26). What is Digital Literacy: Definition and Uses in Daily Life. *Learning.com*. https://www.learning.com/blog/what-is-digital-literacy-definition-and-uses-in-daily-life/

Real, B., & Rose, R. N. (2017). Rural Libraries in the United States: Recent Strides, Future Possibilities, and Meeting Community Needs. *American Library Association*. https://www.ala.org/sites/default/files/advocacy/content/pdfs/Rural%20paper%2007-31-2017.pdf

Socioeconomic status. (n.d.). American Psychological Association. Retrieved March 3, 2026, from https://www.apa.org/topics/socioeconomic-status

State Library of Iowa Continuing Education. (2024, March 27). *We're All Tech Librarians: Digital Literacy* [Video]. YouTube. https://www.youtube.com/watch?v=IAXFODeXBO0

Trollip, A. (n.d.). Understanding the Urban Digital Divide. *Bipartisan Policy Center*. Retrieved October 31, 2024, from https://bipartisanpolicy.org/blog/urban-broadband-blog/

Vogels, E. A. (2020, September 10). 59% of U.S. parents with lower incomes say their child may face digital obstacles in schoolwork. *Pew Research Center*. https://www.pewresearch.org/short-reads/2020/09/10/59-of-u-s-parents-with-lower-incomes-say-their-child-may-face-digital-obstacles-in-schoolwork/

Vogels, E. A. (2021a, June 22). Digital divide persists even as Americans with lower incomes make gains in tech adoption. *Pew Research Center*. https://www.pewresearch.org/short-reads/

2021/06/22/digital-divide-persists-even-as-americans-with-lower-incomes-make-gains-in-tech-adoption/

Vogels, E. A. (2021b, August 17). Despite growth, rural Americans have consistently lower levels of technology ownership than urbanites and lower broadband adoption than suburbanites. *Pew Research Center*. https://www.pewresearch.org/short-reads/2021/08/19/some-digital-divides-persist-between-rural-urban-and-suburban-america/ft_21-06-04_ruralbroadband-png/

What is the "digital divide" and how does your library help? (2024, March 14). *AFPLF*. https://afplf.org/what-is-the-digital-divide/

Yoon, H., Jang, Y., Vaughan, P. W., & Garcia, M. (2018). "Older Adults' Internet Use for Health Information: Digital Divide by Race/Ethnicity and Socioeconomic Status." *Journal of Applied Gerontology, 39*(1), 105-110. https://doi-org.proxy.lib.uiowa.edu/10.1177/0733464818770772